Quantum Fourier Transform

Assessment of the QFT algorithm

ANSWER BOOK

This book accompanies the separate *Question Paper* (80 questions, 3 hours, 240 marks). It contains

1. the **answer key** with the topic tested by every question (page 2), and
2. the **Answer Bank**: every question restated with the correct option marked ✓ and a complete solution with derivations (from page 4).

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

How to evaluate

For each section count C = number of correct answers and W = number of questions answered wrongly (or with more than one bubble darkened). Section marks $= 3C - W$; unanswered questions score 0. The total is out of 240. Every numerical answer was re-computed independently (NumPy, and CUDA-Q state vectors for the QFT circuits) by `verify_answers.py` in the source folder.

Conventions (identical to the Question Paper)

- n qubits, $N = 2^n$, $\omega = e^{2\pi i/N}$. A basis state is written $|x\rangle = |x_1 x_2 \dots x_n\rangle$ with $x = \sum_j x_j 2^{n-j}$: **qubit 1 is the most significant bit** and is written first.
- $\mathrm{QFT}\,|x\rangle = \dfrac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega^{xy}\,|y\rangle, \quad \mathrm{QFT}^{-1}\,|y\rangle = \dfrac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \omega^{-xy}\,|x\rangle$. Binary fraction: $0.x_l x_{l+1} \dots x_m = \frac{x_l}{2} + \frac{x_{l+1}}{4} + \dots + \frac{x_m}{2^{m-l+1}}$.
- $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$; X, Y, Z are the Pauli matrices; $R_k = \mathrm{diag}\left(1, e^{2\pi i/2^k}\right)$, so $R_1 = Z$, $S = R_2$, $T = R_3$; $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- **Textbook QFT circuit:** for $j = 1, \dots, n$: apply H to qubit j, then for $k = 2, \dots, n-j+1$ a controlled-R_k with control qubit $j+k-1$ and target qubit j; finally SWAP qubits j and $n+1-j$ for $j \le n/2$.
- **Phase estimation (QPE):** $U\,|u\rangle = e^{2\pi i\varphi}\,|u\rangle$, $0 \le \varphi < 1$; t counting qubits in $|+\rangle$ control $U^{2^{t-1}}, \dots, U^2, U$; the inverse QFT is applied and the outcome y gives the estimate $\varphi \approx y/2^t$.
- All measurements are in the computational basis unless stated otherwise; log means $\log_2$.

Answer key

Answer distribution: (A) 20, (B) 20, (C) 20, (D) 20.

Section A — Mathematics for the QFT

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
1	**A**	Weighted sum of roots of unity	11	**C**	FFT vs. QFT cost (and its limits)
2	**B**	Orthogonality of roots of unity	12	**A**	Cyclotomic product identity
3	**A**	Binary fractions, periodicity of $e^{2\pi i\theta}$	13	**D**	DFT of a vector; Parseval
4	**D**	Unitarity; the 4×4 Fourier matrix	14	**D**	Binary expansion of $2^k\varphi$
5	**A**	Gauss sum, trace of F_4	15	**C**	Algebra of 8th roots of unity
6	**D**	Interference pattern $\lvert\sum e^{ij\theta}\rvert^2$	16	**B**	Properties of F, F^2, $H\otimes H$, CT
7	**B**	Continued fractions, convergents	17	**C**	Root sums when $\gcd(k,N)>1$
8	**C**	Multiplicative order; Shor's gcd step	18	**C**	Determinants of unitary matrices
9	**B**	Parameters of an n-qubit state	19	**A**	Parseval for the un-normalised DFT
10	**B**	Walsh–Hadamard signs	20	**D**	Spectral leakage ($r\nmid N$)

Section B — Quantum Mechanics

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
21	**B**	Born rule, partial measurement	31	**D**	Commutation of gates
22	**A**	Global vs. relative phase	32	**B**	State-vector memory cost
23	**A**	Interference with a Hadamard	33	**C**	Bell state in the X basis
24	**B**	Bloch-sphere expectation values	34	**B**	Mixture vs. superposition
25	**C**	Entanglement test	35	**C**	Single-qubit gate actions
26	**D**	Phase kickback (CNOT)	36	**D**	Two-qubit correlations
27	**A**	Controlled-phase matrices	37	**C**	Distinguishability; relative phase
28	**B**	Kickback of controlled-U^{2^j}	38	**D**	Product-state criterion
29	**A**	Post-measurement state	39	**D**	Hadamard test
30	**A**	Unitarity of gates	40	**C**	Repetition and confidence

Section C — QFT Basics

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
41	**A**	QFT matrix element ($n=2$)	51	**A**	Constructing the inverse QFT
42	**D**	Output distribution of QFT $\lvert x\rangle$	52	**B**	Phase of a QFT amplitude
43	**C**	Product formula ($n=3$)	53	**B**	QFT vs. NumPy FFT conventions
44	**C**	Controlled-rotation count	54	**D**	QFT $\lvert x\rangle$ is a product state
45	**B**	Two-qubit gates incl. SWAPs	55	**C**	Gate count $n(n-1)/2$
46	**D**	Smallest rotation angle	56	**D**	Two-qubit QFT outputs
47	**A**	Phase on qubit 1; role of SWAPs	57	**A**	Why measuring QFT $\lvert x\rangle$ reveals nothing
48	**C**	QFT^2 = negation	58	**C**	What the QFT does *not* speed up
49	**B**	QFT of the uniform state	59	**B**	Circuit trace: phase build-up
50	**A**	QFT of $\lvert 10\ldots 0\rangle$	60	**D**	Circuit trace: final amplitude, SWAP

Section D — Advanced QFT

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
61	**D**	QFT of a periodic state $(r \mid N)$	71	**D**	Approximate QFT gate savings
62	**B**	Phases of QFT peaks	72	**A**	Periodic state with $r \nmid N$
63	**D**	Shift theorem	73	**D**	Eigenvalues of the QFT
64	**B**	QPE precision vs. counting qubits	74	**C**	Semiclassical QFT
65	**B**	QPE with exact t-bit phase	75	**A**	Applications of the QFT
66	**C**	The $4/\pi^2$ success bound	76	**C**	Period $\leftrightarrow$ peak spacing
67	**D**	QPE distribution for $\varphi = 1/3$	77	**A**	Approximate QFT error bound
68	**A**	Qubit budget of Shor's algorithm	78	**A**	Non-terminating binary phases
69	**B**	Order finding, $a = 7$, $N = 15$	79	**C**	Shor, $N = 21$: continued fractions
70	**B**	Failure mode $\gcd(k, r) > 1$	80	**C**	Shor, $N = 21$: peak probability

Quick-marking grid

1 **A**	2 **B**	3 **A**	4 **D**	5 **A**	6 **D**	7 **B**	8 **C**	9 **B**	10 **B**
11 **C**	12 **A**	13 **D**	14 **D**	15 **C**	16 **B**	17 **C**	18 **C**	19 **A**	20 **D**
21 **B**	22 **A**	23 **A**	24 **B**	25 **C**	26 **D**	27 **A**	28 **B**	29 **A**	30 **A**
31 **D**	32 **B**	33 **C**	34 **B**	35 **C**	36 **D**	37 **C**	38 **D**	39 **D**	40 **C**
41 **A**	42 **D**	43 **C**	44 **C**	45 **B**	46 **D**	47 **A**	48 **C**	49 **B**	50 **A**
51 **A**	52 **B**	53 **B**	54 **D**	55 **C**	56 **D**	57 **A**	58 **C**	59 **B**	60 **D**
61 **D**	62 **B**	63 **D**	64 **B**	65 **B**	66 **C**	67 **D**	68 **A**	69 **B**	70 **B**
71 **D**	72 **A**	73 **D**	74 **C**	75 **A**	76 **C**	77 **A**	78 **A**	79 **C**	80 **C**

Answer Bank — questions with detailed solutions

SECTION A · Mathematics for the QFT Q.1–20

Q.1 SINGLE-CORRECT MCQ · Weighted sum of roots of unity

Let $\omega = e^{2\pi i/8}$. The value of $S = \sum_{k=0}^{7} k\,\omega^k$ is

(A)✓ $\dfrac{8}{\omega - 1}$ **(B)** $\dfrac{8}{1-\omega}$ **(C)** 0 **(D)** $\dfrac{8\omega}{\omega - 1}$

Answer: (A) Solution

Use $\sum_{k=0}^{M-1} kz^k = \dfrac{z(1 - Mz^{M-1} + (M-1)z^M)}{(1-z)^2}$ (differentiate the geometric series $\sum z^k = \frac{1-z^M}{1-z}$ and multiply by z). With $z = \omega$, $M = 8$, $\omega^8 = 1$ and $\omega^7 = \omega^{-1}$:

$$S = \frac{\omega\,(1 - 8\omega^{-1} + 7)}{(1-\omega)^2} = \frac{8\omega - 8}{(1-\omega)^2} = \frac{-8(1-\omega)}{(1-\omega)^2} = \frac{8}{\omega - 1}.$$

Check: $S \neq 0$ because the weights k break the perfect cancellation of $\sum_k \omega^k = 0$; $\frac{8}{1-\omega}$ has the wrong sign. Numerically $S = -4 + 9.657i$.

Q.2 SINGLE-CORRECT MCQ · Orthogonality of roots of unity

The number of ordered pairs (x, y) with $x, y \in \{0, 1, \ldots, 15\}$ for which $\sum_{z=0}^{15} e^{2\pi i\,(x+y)z/16} \neq 0$ is

(A) 1 **(B)✓** 16 **(C)** 15 **(D)** 31

Answer: (B) Solution

The sum is a sum of 16th roots of unity: $\sum_{z=0}^{15} \omega^{kz} = 16$ if $16 \mid k$ and 0 otherwise ($\omega = e^{2\pi i/16}$, geometric series). Here $k = x + y \in \{0, \ldots, 30\}$, so we need $x + y \in \{0, 16\}$. $x + y = 0$: one pair $(0, 0)$. $x + y = 16$: $x = 1, \ldots, 15$ with $y = 16 - x$: 15 pairs. Total $1 + 15 = 16$. (Equivalently: $y \equiv -x \pmod{16}$ has exactly one solution y for every x.)

Q.3 SINGLE-CORRECT MCQ · Binary fractions, periodicity of $e^{2\pi i\theta}$

Let $x = 45$, written with $n = 6$ bits as $x_1 x_2 \ldots x_6$. The value of $e^{2\pi i\,x/2^4}$ is

(A)✓ $e^{i\,13\pi/8}$ **(B)** $e^{i\,5\pi/8}$ **(C)** $e^{i\,3\pi/8}$ **(D)** $e^{-i\,13\pi/8}$

Answer: (A) Solution

$45 = 101101_2$. Dividing by 2^4 moves the binary point four places: $\dfrac{45}{16} = 10.1101_2 = 2 + 0.8125$. Since $e^{2\pi i \cdot 2} = 1$, only the fractional part survives:

$$e^{2\pi i\,45/16} = e^{2\pi i\,0.1101_2} = e^{2\pi i\,\frac{13}{16}} = e^{i\,13\pi/8}.$$

In QFT language: $e^{2\pi i\,x/2^l}$ depends only on the last l bits $0.x_{n-l+1} \ldots x_n$ (here $0.x_3x_4x_5x_6 = 0.1101_2$).

Q.4 SINGLE-CORRECT MCQ · Unitarity; the 4×4 Fourier matrix

The matrix $U = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & c & -1 & d \end{pmatrix}$ is unitary. Then (c, d) equals

(A) $(i,\ -i)$ **(B)** $(1,\ -1)$ **(C)** $(-1,\ 1)$ **(D)** ✓ $(-i,\ i)$

Answer: (D) **Solution**

For a unitary matrix the rows are orthonormal. Let $r_4 = (1, c, -1, d)$.

$$\langle r_1, r_4 \rangle = 1 + c - 1 + d = c + d = 0 \ \Rightarrow\ d = -c.$$

$$\langle r_2, r_4 \rangle = \bar{1} \cdot 1 + \bar{i}\, c + \overline{(-1)}(-1) + \overline{(-i)}\, d = 2 - ic + id = 2 - 2ic = 0 \ \Rightarrow\ c = \tfrac{1}{i} = -i,$$

hence $d = i$. Check $\langle r_3, r_4 \rangle = 1 - c - 1 - d = -(c + d) = 0$ and $\|r_4\|^2 = \frac{1}{4}(1 + 1 + 1 + 1) = 1$. U is exactly the 2-qubit QFT matrix F_4 with entries $i^{xy}/2$ (row 4: $i^{3x} = 1, -i, -1, i$).

Q.5 SINGLE-CORRECT MCQ · Gauss sum, trace of F_4

Let F be the 4×4 unitary Fourier matrix with entries $F_{yx} = \frac{1}{2}\, i^{xy}$ $(x, y = 0, \ldots, 3)$. The trace of F is

(A) ✓ $1 + i$ **(B)** 2 **(C)** 0 **(D)** $1 - i$

Answer: (A) **Solution**

$\operatorname{tr} F = \sum_x F_{xx} = \frac{1}{2}\sum_{x=0}^{3} i^{x^2}$ (a *Gauss sum*). $x^2 = 0, 1, 4, 9$ give $i^0 = 1$, $i^1 = i$, $i^4 = 1$, $i^9 = i$. Hence $\operatorname{tr} F = \frac{1}{2}(2 + 2i) = 1 + i$. Consistency: $F^4 = I$, so its eigenvalues lie in $\{1, -1, i, -i\}$; the eigenvalues of this F are $1, 1, -1, i$ and their sum is $1 + i$.

Q.6 SINGLE-CORRECT MCQ · Interference pattern $|\sum e^{ij\theta}|^2$

Let $f(\theta) = \left| 1 + e^{i\theta} + e^{2i\theta} + e^{3i\theta} \right|^2$. The maximum value of f and the smallest positive θ with $f(\theta) = 0$ are, respectively,

(A) 4 and $\pi/2$ **(B)** 4 and π **(C)** 16 and $\pi/4$ **(D)** ✓ 16 and $\pi/2$

Answer: (D) **Solution**

Geometric series: $\sum_{j=0}^{3} e^{ij\theta} = \dfrac{1 - e^{4i\theta}}{1 - e^{i\theta}} = e^{3i\theta/2}\, \dfrac{\sin 2\theta}{\sin(\theta/2)}$, so

$$f(\theta) = \frac{\sin^2 2\theta}{\sin^2(\theta/2)}.$$

Maximum: all four arrows aligned at $\theta = 0$, $f = |4|^2 = 16$ (by the triangle inequality $f \le 16$). Zeros: $\sin 2\theta = 0$ with $\sin(\theta/2) \neq 0$, i.e. $\theta = \pi/2, \pi, 3\pi/2, \ldots$ The smallest is $\pi/2$ (four arrows pointing in the four directions $1, i, -1, -i$ cancel). This $\sin^2(K\theta/2)/\sin^2(\theta/2)$ shape is the peak shape of every QFT output.

Q.7 SINGLE-CORRECT MCQ · Continued fractions, convergents

The best rational approximation p/q to $\dfrac{11}{64}$ with $1 \le q \le 10$ (as produced by the continued-fraction algorithm) is

(A) 2/9 **(B)** ✓ 1/6 **(C)** 1/7 **(D)** 1/5

Answer: (B) Solution

Continued fraction of $\frac{11}{64}$: $64 = 5\cdot 11 + 9$, $11 = 1\cdot 9 + 2$, $9 = 4\cdot 2 + 1$, $2 = 2\cdot 1$, so $\frac{11}{64} = [0;\,5,\,1,\,4,\,2]$. Convergents: $\frac{0}{1}, \frac{1}{5}, \frac{1}{6}, \frac{5}{29}, \frac{11}{64}$. The last convergent with $q \le 10$ is $\frac{1}{6}$. Direct check: $\left|\frac{11}{64} - \frac{1}{6}\right| = \frac{2}{384} \approx 0.0052$, while $\left|\frac{11}{64} - \frac{1}{5}\right| \approx 0.028$, $\frac{1}{7}$ differs by 0.029 and $\frac{2}{9}$ by 0.050. This is exactly the step that turns a QFT measurement $y = 11$ out of $M = 64$ into the period $r = 6$.

Q.8 SINGLE-CORRECT MCQ · Multiplicative order; Shor's gcd step

The order r of $a = 3$ modulo $N = 35$ (the least $r > 0$ with $a^r \equiv 1 \pmod{35}$), and the numbers $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$, are

(A) $r = 6$; 5 and 7 **(B)** $r = 12$; 1 and 35
(C) ✓ $r = 12$; 7 and 5 **(D)** $r = 4$; 5 and 7

Answer: (C) Solution

Powers of 3 modulo 35: $3, 9, 27, 11, 33, 29, 17, 16, 13, 4, 12, 1$, so $r = 12$ (by Euler, $r \mid \varphi(35) = 24$; $3^6 \equiv 29 \not\equiv 1$ and $3^4 \equiv 11 \not\equiv 1$, so r is neither 6 nor 4). $a^{r/2} = 3^6 \equiv 29 \pmod{35}$, which is not $\equiv -1 = 34$, so Shor's reduction works:

$$\gcd(29 - 1, 35) = \gcd(28, 35) = 7, \qquad \gcd(29 + 1, 35) = \gcd(30, 35) = 5.$$

Indeed $35 = 5 \times 7$.

Q.9 SINGLE-CORRECT MCQ · Parameters of an n-qubit state

A general pure state of 10 qubits is described by D complex amplitudes. After removing normalisation and the unobservable global phase, the number of independent *real* parameters is P. Then (D, P) is

(A) (1024, 2048) **(B)** ✓ (1024, 2046) **(C)** (1024, 2047) **(D)** (20, 40)

Answer: (B) Solution

n qubits span a 2^n-dimensional space: $D = 2^{10} = 1024$ complex numbers = 2048 real numbers. Normalisation $\sum |\alpha_x|^2 = 1$ removes one real parameter and the global phase $e^{i\gamma}$ removes another: $P = 2048 - 2 = 2046$. (Option $(1024, 2047)$ forgets the global phase; $(20, 40)$ counts as if qubits were independent.)

Q.10 Single-correct MCQ · Walsh–Hadamard signs

For $n = 4$, $H^{\otimes 4}|x\rangle = \frac{1}{4}\sum_y (-1)^{x\cdot y}|y\rangle$ where $x \cdot y = \sum_j x_j y_j$. For $x = 1011_2$, the number of $y \in \{0, \ldots, 15\}$ with a *negative* amplitude is

(A) 4 **(B)** ✓ 8 **(C)** 12 **(D)** 3

Answer: (B) Solution

The amplitude is negative iff $x \cdot y = y_1 + y_3 + y_4$ is odd (because $x = 1011$ has $x_1 = x_3 = x_4 = 1$). y_2 is free (2 choices) and among the 8 choices of (y_1, y_3, y_4) exactly half have odd sum, so the count is $2 \times 4 = 8$. In general, for any $x \neq 0$ exactly 2^{n-1} of the 2^n signs are negative (the characters $(-1)^{x\cdot y}$ of $\mathbb{Z}_2^n$ sum to zero).

Q.11 Single-correct MCQ · FFT vs. QFT cost (and its limits)

A classical FFT on a vector of length 2^n needs about $n\,2^n$ operations, while the QFT circuit needs $\frac{n(n+1)}{2} + \lfloor n/2 \rfloor$ gates. For $n = 20$ the ratio (FFT operations)/(QFT gates) is closest to

(A) 10^9 **(B)** 10^3 **(C)** ✓ 10^5 **(D)** 10^7

Answer: (C) Solution

FFT: $20 \times 2^{20} = 20 \times 1\,048\,576 = 20\,971\,520$. QFT: $\frac{20\cdot 21}{2} + 10 = 210 + 10 = 220$ gates. Ratio $= 20\,971\,520/220 \approx 9.5 \times 10^4 \approx 10^5$. Caveat (important for interpretation): the QFT output is stored in amplitudes and cannot be read out completely, so this ratio is *not* a speed-up for computing all coefficients.

Q.12 Single-correct MCQ · Cyclotomic product identity

With $\omega = e^{2\pi i/8}$, the value of $\displaystyle\prod_{k=1}^{7} |1 - \omega^k|$ is

(A) ✓ 8 **(B)** 0 **(C)** 1 **(D)** 7

Answer: (A) Solution

The roots of $z^8 - 1$ are $\omega^0, \ldots, \omega^7$, so $z^8 - 1 = \prod_{k=0}^{7}(z - \omega^k)$ and

$$\frac{z^8 - 1}{z - 1} = 1 + z + \cdots + z^7 = \prod_{k=1}^{7}(z - \omega^k).$$

Put $z = 1$: $\prod_{k=1}^{7}(1 - \omega^k) = 8$, hence the product of the moduli is 8. (Geometrically: the product of the distances from one vertex of a regular octagon inscribed in the unit circle to the other seven vertices.)

Q.13 Single-correct MCQ · DFT of a vector; Parseval

The normalised vector $\alpha = \frac{1}{3}(1,2,2,0)$ is transformed by the unitary 4×4 Fourier matrix $\beta_y = \frac{1}{2}\sum_x \alpha_x\, i^{xy}$. The value of $|\beta_0|^2$ is

(A) $\frac{5}{6}$ **(B)** $\frac{25}{12}$ **(C)** $\frac{1}{4}$ **(D)** ✓ $\frac{25}{36}$

Answer: (D) Solution

$\beta_0 = \frac{1}{2}\sum_x \alpha_x = \frac{1}{2}\cdot\frac{1+2+2+0}{3} = \frac{5}{6}$, so $|\beta_0|^2 = \frac{25}{36}$. Check the other options: $\frac{5}{6}$ is the amplitude, not the probability; $\frac{25}{12} > 1$ is impossible because $\sum_y |\beta_y|^2 = \sum_x |\alpha_x|^2 = 1$ (Parseval, F is unitary). The remaining probability $\frac{11}{36}$ is spread over $y = 1,2,3$.

Q.14 Single-correct MCQ · Binary expansion of $2^k\varphi$

Let $\varphi = 0.\varphi_1\varphi_2\varphi_3$ (binary, $\varphi_j \in \{0,1\}$). Then $e^{2\pi i\, 4\varphi}$ equals

(A) $e^{2\pi i\, 0.\varphi_2\varphi_3}$ **(B)** i^{φ_2} **(C)** $(-1)^{\varphi_1}$ **(D)** ✓ $(-1)^{\varphi_3}$

Answer: (D) Solution

$4\varphi = \varphi_1\varphi_2.\varphi_3$ in binary, i.e. $4\varphi = (2\varphi_1+\varphi_2)+0.\varphi_3$. The integer part is a whole number of turns, so $e^{2\pi i\, 4\varphi} = e^{2\pi i\, 0.\varphi_3} = e^{i\pi\varphi_3} = (-1)^{\varphi_3}$. This is why, in phase estimation, the counting qubit controlling U^4 carries only the *last* bit φ_3, and the inverse QFT reads the bits from there.

Q.15 Matching lists · Algebra of 8th roots of unity

Let $\omega = e^{2\pi i/8}$. Match List-I with List-II.

List-I		List-II	
(P)	ω^4	**(1)**	$\sqrt{2}$
(Q)	$\sum_{k=0}^{7}\omega^{2k}$	**(2)**	-1
(R)	$\omega\cdot\omega^7$	**(3)**	0
(S)	$\omega+\omega^7$	**(4)**	1

(A) P→1, Q→3, R→4, S→2 **(B)** P→2, Q→4, R→3, S→1

(C) ✓ P→2, Q→3, R→4, S→1 **(D)** P→3, Q→2, R→1, S→4

Answer: (C) Solution

P: $\omega^4 = e^{i\pi} = -1$ (2). Q: $\omega^2 = i$ is a primitive 4th root, and $\sum_{k=0}^{7} i^k = 2(1+i-1-i) = 0$ (3). R: $\omega\,\omega^7 = \omega^8 = 1$ (4). S: $\omega+\omega^7 = \omega+\bar{\omega} = 2\cos\frac{\pi}{4} = \sqrt{2}$ (1). Hence P→2, Q→3, R→4, S→1.

Q.16 Matching lists · Properties of F, F^2, $H \otimes H$, CT

F is the 4×4 unitary Fourier matrix ($F_{yx} = i^{xy}/2$), H the Hadamard matrix and $CT = \mathrm{diag}(1, 1, 1, e^{i\pi/4})$. Match each matrix in List-I with the *only* property in List-II that it has.

List-I		List-II	
(P)	F	**(1)**	a permutation matrix that is its own inverse
(Q)	F^2	**(2)**	real, symmetric, its own inverse, but not a permutation matrix
(R)	$H \otimes H$	**(3)**	diagonal with determinant $e^{i\pi/4}$
(S)	CT	**(4)**	$M^4 = I$ but $M^2 \neq I$

(A) P→4, Q→2, R→1, S→3
(B) ✓ P→4, Q→1, R→2, S→3
(C) P→3, Q→1, R→2, S→4
(D) P→1, Q→4, R→2, S→3

Answer: (B) Solution

$(F^2)_{zx} = \frac{1}{4}\sum_y i^{y(x+z)} = \delta_{z,\,-x \bmod 4}$, so F^2 is the permutation $|x\rangle \mapsto |-x\rangle$ ($0 \to 0$, $1 \leftrightarrow 3$, $2 \to 2$), which squares to I: Q→1. Hence $F^4 = I$ while $F^2 \neq I$: P→4. $H \otimes H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$ is real, symmetric and $(H \otimes H)^2 = I$, but its entries are $\pm\frac{1}{2}$, so it is not a permutation: R→2. CT is diagonal with $\det = e^{i\pi/4}$; note $CT^4 = \mathrm{diag}(1, 1, 1, -1) \neq I$, so it does not have property 4: S→3.

Q.17 Assertion–Reason · Root sums when $\gcd(k, N) > 1$

Assertion (A): For every integer k that is not a multiple of N, $\displaystyle\sum_{j=0}^{N-1} e^{2\pi i\, jk/N} = 0$.

Reason (R): For every integer k that is not a multiple of N, the N numbers $e^{2\pi i\, jk/N}$ ($j = 0, \ldots, N-1$) are the N distinct vertices of a regular N-gon centred at the origin.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) ✓ A is true, but R is false
(D) A is false, but R is true

Answer: (C) Solution

A is true: with $z = e^{2\pi ik/N} \neq 1$, $\sum_{j=0}^{N-1} z^j = \frac{1-z^N}{1-z} = \frac{1-e^{2\pi ik}}{1-z} = 0$. **R is false:** if $g = \gcd(k, N) > 1$ the numbers repeat. Example $N = 8$, $k = 2$: the values are $1, i, -1, -i, 1, i, -1, -i$ — a square traversed twice, not 8 distinct vertices of an octagon. (The sum is still zero, because a regular N/g-gon is traversed g times.) Answer: (C).

Q.18 Statement I / II · Determinants of unitary matrices

Statement I: Every unitary matrix U satisfies $|\det U| = 1$.

Statement II: The unitary Fourier matrix F_N ($N = 2^n$, entries $\omega^{xy}/\sqrt{N}$) has $\det F_N = \pm 1$ for every n.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Solution

I is true: $|\det U|^2 = \det(U^\dagger U) = \det I = 1$. **II is false:** the determinant is the product of the eigenvalues, which lie in $\{1, -1, i, -i\}$ because $F^4 = I$. For $N = 4$ the eigenvalues are $1, 1, -1, i$ (compare the trace $1 + i$ in Q.5), so $\det F_4 = 1 \cdot 1 \cdot (-1) \cdot i = -i \neq \pm 1$. (For $N = 2$, $F_2 = H$ and $\det H = -1$, so the claim happens to hold there — one counterexample is enough.) Answer: (C).

Paragraph for Questions 19 and 20

Let $N = 32$ and define the sequence $a_x = 1$ if $x \equiv 2 \pmod 5$ and $a_x = 0$ otherwise, for $x = 0, 1, \ldots, 31$. Its (un-normalised) Fourier coefficients are $A_y = \sum_{x=0}^{31} a_x\, e^{2\pi i\, xy/32}$, $y = 0, 1, \ldots, 31$. (The sequence is "periodic with period 5", but 5 does not divide 32.)

Q.19 Paragraph based · Parseval for the un-normalised DFT

The value of $\displaystyle\sum_{y=0}^{31} |A_y|^2$ is

(A) ✓ 192 **(B)** 6 **(C)** 36 **(D)** 1152

Answer: (A) Solution

The non-zero samples are $x = 2, 7, 12, 17, 22, 27$, i.e. $K = 6$ ones. For the un-normalised transform $A = \sqrt{N}\, Fa$ with unitary F, Parseval gives

$$\sum_y |A_y|^2 = N \sum_x |a_x|^2 = 32 \times 6 = 192.$$

(36 is $|A_0|^2$ alone; 6 is $\sum_x |a_x|^2$; 1152 would be $N \cdot 36$.)

Q.20 Paragraph based · Spectral leakage ($r \nmid N$)

Among $y = 1, 2, \ldots, 15$, the value of y for which $|A_y|$ is largest is

(A) 6

(B) 16 is the only peak; all of $1, \ldots, 15$ give $|A_y| = 0$

(C) 7

(D) ✓ 13

Answer: (D) Solution

$A_y = \sum_{j=0}^{5} e^{2\pi i(5j+2)y/32} = e^{4\pi i y/32} \sum_{j=0}^{5} e^{2\pi i\, 5jy/32}$, so

$$|A_y| = \left|\frac{\sin(6 \cdot 5\pi y/32)}{\sin(5\pi y/32)}\right| = \left|\frac{\sin(30\pi y/32)}{\sin(5\pi y/32)}\right|.$$

Peaks occur where $5y/32$ is close to an integer, i.e. $y \approx 32k/5 = 6.4,\ 12.8,\ \ldots$ The closer $5y/32$ is to an integer, the higher the peak: $5 \cdot 13/32 = 2.03$ (distance 0.03) beats $5 \cdot 6/32 = 0.94$ (distance 0.06) and $5 \cdot 7/32 = 1.09$. Numerically $|A_{13}| = 5.67$, $|A_6| = 4.74$, $|A_7| = 3.38$, $|A_{12}| = 1.85$; and $|A_{16}| = 0$ since $\sin(15\pi) = 0$. Hence $y = 13$. Lesson: with a period that does not divide N, the QFT peaks "leak" and their heights depend on how close ky/N is to an integer.

SECTION B · Quantum Mechanics Q.21–40

Q.21 Single-correct MCQ · Born rule, partial measurement

A two-qubit state is $|\psi\rangle = c\,(\,|00\rangle + 2i\,|01\rangle - |10\rangle + \sqrt{2}\,|11\rangle\,)$ with $c > 0$. If only the *first* qubit is measured in the computational basis, the probability of obtaining 1 is

(A) $\frac{1}{2}$ **(B)** ✓ $\frac{3}{8}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{5}{8}$

Answer: (B) Solution

Normalisation: $c^2(|1|^2 + |2i|^2 + |-1|^2 + |\sqrt{2}|^2) = c^2(1+4+1+2) = 8c^2 = 1$, so $c = \frac{1}{2\sqrt{2}}$. By the Born rule, the probability that the first qubit gives 1 is the total weight of the basis states whose first bit is 1:

$$P(1) = c^2(|-1|^2 + |\sqrt{2}|^2) = \frac{1+2}{8} = \frac{3}{8}.$$

(The phases i and -1 do not affect probabilities in this basis; $\frac{5}{8}$ is $P(0)$; $\frac{1}{4}$ and $\frac{1}{2}$ come from forgetting to normalise or from counting states instead of weights.)

Q.22 Single-correct MCQ · Global vs. relative phase

Which pair of single-qubit state vectors describes the *same* physical state (i.e. gives identical measurement statistics in every basis)?

(A) ✓ $|0\rangle + i\,|1\rangle$ and $i\,|0\rangle - |1\rangle$
(B) $|0\rangle + i\,|1\rangle$ and $i\,|0\rangle + |1\rangle$
(C) $|0\rangle + i\,|1\rangle$ and $|0\rangle - i\,|1\rangle$
(D) $|0\rangle + |1\rangle$ and $|0\rangle - |1\rangle$

Answer: (A) Solution

Two vectors represent the same physical state iff they differ by an overall (global) factor $e^{i\gamma}$. $i\,|0\rangle - |1\rangle = i(\,|0\rangle + i\,|1\rangle\,)$, since $i \cdot i = -1$: global phase i, so the pair is physically identical. The other pairs differ by a *relative* phase: $|0\rangle \pm i\,|1\rangle$ (relative phase $\pm i$), $|0\rangle \pm |1\rangle$ (relative phase ± 1), and $i\,|0\rangle + |1\rangle = i(|0\rangle - i\,|1\rangle)$, which is $|0\rangle - i\,|1\rangle$ again. A relative phase is measurable: e.g. $|0\rangle + |1\rangle \propto |+\rangle$ and $|0\rangle - |1\rangle \propto |-\rangle$ are perfectly distinguishable in the X basis. Relative phases are exactly what the QFT writes information into.

Q.23 Single-correct MCQ · Interference with a Hadamard

The state $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{i\phi}\,|1\rangle\,)$ passes through a Hadamard gate and is then measured in the computational basis. The probability of outcome 0 is

(A) ✓ $\cos^2(\phi/2)$ **(B)** $\sin^2(\phi/2)$ **(C)** $\frac{1}{2}$ **(D)** $\cos^2\phi$

Answer: (A) Solution

With $H\,|0\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $H\,|1\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$:

$$H\,\frac{|0\rangle + e^{i\phi}\,|1\rangle}{\sqrt{2}} = \frac{1+e^{i\phi}}{2}\,|0\rangle + \frac{1-e^{i\phi}}{2}\,|1\rangle\,.$$

$|1+e^{i\phi}|^2 = (1+e^{i\phi})(1+e^{-i\phi}) = 2 + 2\cos\phi = 4\cos^2(\phi/2)$, so $P(0) = \frac{4\cos^2(\phi/2)}{4} = \cos^2(\phi/2)$ (and $P(1) = \sin^2(\phi/2)$). The Hadamard converts an invisible *relative phase* into a visible *probability*: $\phi = 0$ gives 0 surely, $\phi = \pi$ gives 1 surely. A single-qubit QFT is exactly this H.

Q.24 SINGLE-CORRECT MCQ · Bloch-sphere expectation values

For the qubit state $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$ the quantity $\langle X\rangle + i\langle Y\rangle$, where $\langle A\rangle = \langle\psi|A|\psi\rangle$ and X, Y are Pauli matrices, equals

(A) $\cos\theta\, e^{i\phi}$ **(B)** ✓ $\sin\theta\, e^{i\phi}$ **(C)** $\sin\frac{\theta}{2}\, e^{i\phi}$ **(D)** $\sin\theta\, e^{-i\phi}$

Answer: (B) Solution

Write $a = \cos\frac{\theta}{2}$, $b = e^{i\phi}\sin\frac{\theta}{2}$. With $X = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, $Y = \left(\begin{smallmatrix}0&-i\\i&0\end{smallmatrix}\right)$:

$$\langle X\rangle = \bar{a}b + \bar{b}a = 2\,\mathrm{Re}(\bar{a}b), \qquad \langle Y\rangle = -i\bar{a}b + i\bar{b}a = 2\,\mathrm{Im}(\bar{a}b).$$

Hence $\langle X\rangle + i\langle Y\rangle = 2\bar{a}b = 2\cos\frac{\theta}{2}\sin\frac{\theta}{2}\,e^{i\phi} = \sin\theta\, e^{i\phi}$. So $\langle X\rangle = \sin\theta\cos\phi$, $\langle Y\rangle = \sin\theta\sin\phi$ (and $\langle Z\rangle = \cos\theta$): the Bloch vector $(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. The *azimuth* ϕ is the phase that QFT rotations change.

Q.25 SINGLE-CORRECT MCQ · Entanglement test

Which of the following two-qubit states is *entangled* (cannot be written as $|a\rangle \otimes |b\rangle$)?

(A) $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$ **(B)** $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$
(C) ✓ $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$ **(D)** $\frac{1}{2}(|00\rangle + i|01\rangle - |10\rangle - i|11\rangle)$

Answer: (C) Solution

For $\sum c_{jk}|jk\rangle$, a product $(a_0|0\rangle + a_1|1\rangle)(b_0|0\rangle + b_1|1\rangle)$ has $c_{jk} = a_j b_k$, so $c_{00}c_{11} = a_0a_1b_0b_1 = c_{01}c_{10}$. Conversely, $c_{00}c_{11} = c_{01}c_{10}$ (the 2×2 coefficient matrix has determinant 0, i.e. rank 1) guarantees a product. Test each option (factor $\frac{1}{2}$ dropped):

- $(1, 1, 1, -1)$: $c_{00}c_{11} = -1 \neq 1 = c_{01}c_{10} \Rightarrow$ **entangled**.
- $(1, 1, 1, 1)$: $1 = 1$, product $|+\rangle|+\rangle$.
- $(1, -1, 1, -1)$: $-1 = -1$, product $|+\rangle|-\rangle$.
- $(1, i, -1, -i)$: $-i = -i$, product $\frac{1}{2}(|0\rangle - |1\rangle)(|0\rangle + i|1\rangle)$.

The entangled state is $\mathrm{CZ}|+\rangle|+\rangle$: a controlled-phase gate (like those in the QFT) *can* entangle. (This is not in conflict with Q.54: the QFT acting on a *basis* state happens to give a product state.)

Q.26 SINGLE-CORRECT MCQ · Phase kickback (CNOT)

A CNOT gate (qubit 1 = control, qubit 2 = target) acts on $|+\rangle|-\rangle$, where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. The output is

(A) $|+\rangle|-\rangle$ **(B)** $|-\rangle|+\rangle$
(C) $\frac{1}{\sqrt{2}}(|0\rangle|-\rangle + |1\rangle|+\rangle)$ **(D)** ✓ $|-\rangle|-\rangle$

Answer: (D) Solution

$|-\rangle$ is an eigenvector of X with eigenvalue -1: $X|-\rangle = \frac{|1\rangle - |0\rangle}{\sqrt{2}} = -|-\rangle$. Therefore

$$\mathrm{CNOT}\,|+\rangle|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle|-\rangle + |1\rangle\, X|-\rangle) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|-\rangle = |-\rangle|-\rangle.$$

The target is unchanged; the eigenvalue -1 has been "kicked back" onto the control as a relative phase. This *phase kickback* is the mechanism by which controlled-U^{2^j} gates load the eigenphase into the counting register in phase estimation (see Q.28).

Q.27 SINGLE-CORRECT MCQ · Controlled-phase matrices

In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (first qubit = control), the matrix of controlled-R_3, where $R_k = \mathrm{diag}\,(1, e^{2\pi i/2^k})$, is

(A) ✓ $\mathrm{diag}\,(1,1,1,e^{i\pi/4})$ **(B)** $\mathrm{diag}\,(1,e^{i\pi/4},1,e^{i\pi/4})$
(C) $\mathrm{diag}\,(1,1,e^{i\pi/4},e^{i\pi/4})$ **(D)** $\mathrm{diag}\,(1,1,1,e^{i\pi/8})$

Answer: (A) Solution

$R_3 = \mathrm{diag}(1, e^{2\pi i/8}) = \mathrm{diag}(1, e^{i\pi/4})$ (the T gate). Controlled-R_3 applies R_3 to the target only when the control is $|1\rangle$:

$$\mathrm{C}R_3 = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes R_3 = \mathrm{diag}(1,1,1,e^{i\pi/4}).$$

Only $|11\rangle$ picks up a phase, so the gate is *symmetric*: it does not matter which qubit is called the control. $\mathrm{diag}(1,e^{i\pi/4},1,e^{i\pi/4}) = I \otimes R_3$ and $\mathrm{diag}(1,1,e^{i\pi/4},e^{i\pi/4}) = R_3 \otimes I$ are uncontrolled; $e^{i\pi/8}$ confuses $2\pi/2^3$ with $\pi/2^3$.

Q.28 SINGLE-CORRECT MCQ · Kickback of controlled-U^{2^j}

$U|u\rangle = e^{2\pi i\varphi}|u\rangle$. A control qubit in $|+\rangle$ and the register in $|u\rangle$ pass through a controlled-U^{2^j} gate. The output state is

(A) $\frac{1}{\sqrt{2}}\, e^{2\pi i\, 2^j\varphi}(\,|0\rangle + |1\rangle\,)\,|u\rangle$ **(B)** ✓ $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\, 2^j\varphi}\,|1\rangle\,)\,|u\rangle$
(C) $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\, j\varphi}\,|1\rangle\,)\,|u\rangle$ **(D)** $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\,\varphi/2^j}\,|1\rangle\,)\,|u\rangle$

Answer: (B) Solution

$U^{2^j}|u\rangle = (e^{2\pi i\varphi})^{2^j}|u\rangle = e^{2\pi i\, 2^j\varphi}|u\rangle$. The gate acts only on the $|1\rangle$ branch:

$$\frac{1}{\sqrt{2}}(\,|0\rangle\,|u\rangle + |1\rangle\; U^{2^j}|u\rangle\,) = \frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\, 2^j\varphi}\,|1\rangle\,)\,|u\rangle\,.$$

The phase is *relative* (only on $|1\rangle$), not global, so it is measurable after a Hadamard / inverse QFT. If $\varphi = 0.\varphi_1\varphi_2\ldots$ in binary, $2^j\varphi$ shifts the binary point j places and the phase is $e^{2\pi i\, 0.\varphi_{j+1}\varphi_{j+2}\cdots}$ — exactly the factor in the QFT product formula.

Q.29 SINGLE-CORRECT MCQ · Post-measurement state

The state $\frac{1}{\sqrt{3}}(\,|00\rangle + |01\rangle + |11\rangle\,)$ is prepared and only the *first* qubit is measured, giving 0. The probability of this outcome and the post-measurement state of the second qubit are

(A) ✓ $\frac{2}{3}$ and $|+\rangle$ **(B)** $\frac{1}{3}$ and $|0\rangle$ **(C)** $\frac{2}{3}$ and $|0\rangle$ **(D)** $\frac{1}{3}$ and $|+\rangle$

Answer: (A) Solution

Group by the first qubit: $\frac{1}{\sqrt{3}}\Big(\,|0\rangle\,(\,|0\rangle + |1\rangle\,) + |1\rangle\,|1\rangle\,\Big) = \sqrt{\frac{2}{3}}\,|0\rangle\,|+\rangle + \sqrt{\frac{1}{3}}\,|1\rangle\,|1\rangle$. Probability of first qubit = 0: $\left|\sqrt{2/3}\right|^2 = \frac{2}{3}$. By the projection postulate the state collapses to the normalised $|0\rangle\,|+\rangle$, so the second qubit is $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. (Partial measurement keeps the *relative* amplitudes of the surviving branch; this is why in Shor's algorithm measuring the work register leaves the counting register in a periodic superposition.)

Q.30 Single-correct MCQ · Unitarity of gates

Which of the following matrices can *not* be a quantum gate?

(A) ✓ $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ i & i\end{pmatrix}$ **(B)** $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$ **(C)** $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & i\\ i & 1\end{pmatrix}$ **(D)** $\begin{pmatrix}0 & 1\\ i & 0\end{pmatrix}$

Answer: (A) Solution

Closed-system evolution is unitary: $U^\dagger U = I$, i.e. the columns are orthonormal. For $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ i & i\end{smallmatrix}\right)$ both columns equal $\frac{1}{\sqrt{2}}(1,i)^T$, so their inner product is $1 \neq 0$ (also det $= 0$): it maps $|0\rangle$ and $|1\rangle$ to the same state and would destroy information — not unitary. Checks: H is unitary; for $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & i\\ i & 1\end{smallmatrix}\right)$ the column inner product is $\frac{1}{2}(\bar{1}\cdot i + \bar{i}\cdot 1) = \frac{1}{2}(i-i) = 0$ and each column has norm 1; $\left(\begin{smallmatrix}0 & 1\\ i & 0\end{smallmatrix}\right)$ has columns $(0,i)$ and $(1,0)$, orthonormal. (Unitarity is also why the QFT is invertible: $F^{-1} = F^\dagger$.)

Q.31 Single-correct MCQ · Commutation of gates

Which two-qubit gate does *not* commute with controlled-$R_2 = \mathrm{diag}(1,1,1,i)$?

(A) $Z \otimes Z$ **(B)** $S \otimes I$, where $S = \mathrm{diag}(1,i)$

(C) SWAP **(D)** ✓ $H \otimes I$

Answer: (D) Solution

$Z \otimes Z$ and $S \otimes I$ are diagonal; diagonal matrices always commute. SWAP exchanges $|01\rangle \leftrightarrow |10\rangle$; CR_2 has the same entry (1) on both, so SWAP $\cdot\, CR_2 \cdot$ SWAP $= CR_2$: they commute (controlled-phase gates are symmetric). $H \otimes I$ mixes $|0x\rangle$ with $|1x\rangle$. For $x = 1$ the diagonal entries of CR_2 on $|01\rangle$ and $|11\rangle$ are 1 and i — different — so the two operators do not commute. Explicitly $(H \otimes I)\, CR_2\, |01\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$ but $CR_2 (H \otimes I)\, |01\rangle = \frac{1}{\sqrt{2}}(|01\rangle + i\,|11\rangle)$. This is why the order "H first, then the controlled rotations" in the QFT circuit matters.

Q.32 Single-correct MCQ · State-vector memory cost

A state-vector simulator stores each amplitude as a single-precision complex number (two 4-byte floats). The memory needed for the full state of 32 qubits is

(A) 256 bytes **(B)** ✓ 32 GiB **(C)** 16 GiB **(D)** 64 GiB

Answer: (B) Solution

n qubits $\Rightarrow 2^n$ amplitudes. Each amplitude occupies $2 \times 4 = 8$ bytes. Memory $= 2^{32} \times 8 = 2^{35}$ bytes $= 2^5 \times 2^{30}$ bytes $= 32$ GiB. Double precision (16 bytes) would need 64 GiB; 256 bytes (32×8) is the wrong idea that each qubit needs one number. Every extra qubit doubles the memory — the reason the course notebooks default to 15 qubits ($2^{15} \times 8$ B $= 256$ KiB) on a laptop.

Q.33 Single-correct MCQ · Bell state in the X basis

$H \otimes H$ is applied to the Bell state $\frac{1}{\sqrt{2}}(\,|00\rangle - |11\rangle\,)$, followed by a measurement of both qubits. The probability of the outcome 01 is

(A) 0 **(B)** $\frac{1}{4}$ **(C)** ✓ $\frac{1}{2}$ **(D)** 1

Answer: (C) Solution

$H \otimes H\,|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ and $H \otimes H\,|11\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$ (sign $(-1)^{x\cdot y}$). Subtract and divide by $\sqrt{2}$:

$$\frac{1}{2\sqrt{2}}(2\,|01\rangle + 2\,|10\rangle\,) = \frac{1}{\sqrt{2}}(\,|01\rangle + |10\rangle\,).$$

Hence $P(01) = P(10) = \frac{1}{2}$ and $P(00) = P(11) = 0$. The perfect "equal bits" correlation in the Z basis became a perfect "opposite bits" correlation in the X basis — interference of four amplitudes, not a product-state result ($\frac{1}{4}$).

Q.34 Single-correct MCQ · Mixture vs. superposition

A source emits a qubit that is $|0\rangle$ with probability $\frac{1}{2}$ and $|1\rangle$ with probability $\frac{1}{2}$ (a classical coin decides; nobody knows which). A Hadamard gate is applied and the qubit is measured. The probability of outcome 0 is

(A) 1 **(B)** ✓ $\frac{1}{2}$ **(C)** $\frac{1}{4}$ **(D)** 0

Answer: (B) Solution

This is a *mixture*, not a superposition. Treat the two cases separately and average: $H\,|0\rangle = |+\rangle$ gives $P(0) = \frac{1}{2}$; $H\,|1\rangle = |-\rangle$ gives $P(0) = \frac{1}{2}$. Hence $P(0) = \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{2}$. (Density matrix: $\rho = \frac{1}{2}I$, and $H\rho H^\dagger = \frac{1}{2}I$.) Contrast: the superposition $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ gives $H\,|+\rangle = |0\rangle$ and $P(0) = 1$. Amplitudes interfere, classical probabilities do not — QFT peaks need coherence, which is why decoherence (dephasing) destroys them.

Q.35 Matching lists · Single-qubit gate actions

Match each single-qubit gate in List-I with its action in List-II. ($R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$.)

List-I		List-II	
(P)	X	**(1)**	$\lvert1\rangle \mapsto \frac{1}{\sqrt{2}}(\,\lvert0\rangle - \lvert1\rangle\,)$
(Q)	H	**(2)**	$\lvert1\rangle \mapsto e^{i\pi/4}\,\lvert1\rangle$
(R)	R_2	**(3)**	$\lvert0\rangle \leftrightarrow \lvert1\rangle$
(S)	R_3	**(4)**	$\lvert1\rangle \mapsto i\,\lvert1\rangle$

(A) P→3, Q→1, R→2, S→4 **(B)** P→1, Q→3, R→4, S→2

(C) ✓ P→3, Q→1, R→4, S→2 **(D)** P→4, Q→1, R→3, S→2

Answer: (C) Solution

X is the bit flip: $|0\rangle \leftrightarrow |1\rangle$ (3). $H\,|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$ (1). $R_2 = \mathrm{diag}(1, e^{2\pi i/4}) = \mathrm{diag}(1, i) = S$, so $|1\rangle \mapsto i\,|1\rangle$ (4). $R_3 = \mathrm{diag}(1, e^{2\pi i/8}) = T$, so $|1\rangle \mapsto e^{i\pi/4}\,|1\rangle$ (2). Hence P→3, Q→1, R→4, S→2. Note $R_1 = Z$, $R_2 = S$, $R_3 = T$, and $R_k^2 = R_{k-1}$.

Q.36 Matching lists · Two-qubit correlations

Match each two-qubit state in List-I with its statistics when *both* qubits are measured in the computational basis (List-II).

List-I		List-II	
(P)	$\frac{1}{\sqrt{2}}(\,\lvert 00\rangle + \lvert 11\rangle\,)$	**(1)**	all four outcomes, each with probability $\frac{1}{4}$
(Q)	$\frac{1}{\sqrt{2}}(\,\lvert 01\rangle - \lvert 10\rangle\,)$	**(2)**	the two bits are always equal; 00 and 11 each $\frac{1}{2}$
(R)	$\frac{1}{2}(\,\lvert 0\rangle + i\,\lvert 1\rangle\,)(\,\lvert 0\rangle - \lvert 1\rangle\,)$	**(3)**	first bit always 0; second bit 0 or 1 with $\frac{1}{2}$ each
(S)	$\lvert 0\rangle\,\lvert +\rangle$	**(4)**	the two bits always differ; 01 and 10 each $\frac{1}{2}$

(A) P→1, Q→4, R→2, S→3 **(B)** P→4, Q→2, R→1, S→3

(C) P→2, Q→4, R→3, S→1 **(D)** ✓ P→2, Q→4, R→1, S→3

Answer: (D) Solution

P (Φ^+): amplitudes only on $|00\rangle$, $|11\rangle$, each $\frac{1}{\sqrt{2}}$ ⇒ equal bits, $\frac{1}{2}$ each (2). Q (Ψ^-): only $|01\rangle$, $|10\rangle$ ⇒ bits always differ (4); the minus sign is invisible in this basis. R: expand $\frac{1}{2}(|00\rangle - |01\rangle + i\,|10\rangle - i\,|11\rangle)$; every amplitude has modulus $\frac{1}{2}$ ⇒ all four outcomes with probability $\frac{1}{4}$ (1). The phases i and -1 do not show up — just like the output of a QFT on a basis state. S: $|0\rangle\,|+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$ ⇒ first bit 0, second random (3). Hence P→2, Q→4, R→1, S→3.

Q.37 Assertion–Reason · Distinguishability; relative phase

Assertion (A): The states $|+\rangle$ and $|-\rangle$ cannot be distinguished by a measurement in the computational basis $\{|0\rangle, |1\rangle\}$.

Reason (R): $|+\rangle$ and $|-\rangle$ differ only by a global phase.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) ✓ A is true, but R is false

(D) A is false, but R is true

Answer: (C) Solution

A is true: for both states $P(0) = P(1) = \frac{1}{2}$, because $|\pm\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$. **R is false:** $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ is not $e^{i\gamma}\,|+\rangle$ for any γ; the two differ by a *relative* phase -1 on $|1\rangle$. They are orthogonal ($\langle +|-\rangle = 0$) and therefore perfectly distinguishable in the X basis (apply H, then measure: $|+\rangle \to |0\rangle$, $|-\rangle \to |1\rangle$). States that differ by a global phase could never be distinguished in *any* basis. Answer: (C).

Q.38 Statement I / II · Product-state criterion

Statement I: The Bell state $\frac{1}{\sqrt{2}}(\,|00\rangle + |11\rangle\,)$ can be written as $|a\rangle \otimes |b\rangle$ for suitable single-qubit states $|a\rangle$, $|b\rangle$.

Statement II: A two-qubit state $c_{00}\,|00\rangle + c_{01}\,|01\rangle + c_{10}\,|10\rangle + c_{11}\,|11\rangle$ is a product state if and only if $c_{00}c_{11} = c_{01}c_{10}$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) ✓ Statement I is false, but Statement II is true

Answer: (D) Solution

II is true. ($\Rightarrow$) If the state is $(a_0\,|0\rangle + a_1\,|1\rangle)(b_0\,|0\rangle + b_1\,|1\rangle)$ then $c_{jk} = a_j b_k$ and $c_{00}c_{11} = a_0b_0a_1b_1 = c_{01}c_{10}$. ($\Leftarrow$) If $c_{00}c_{11} = c_{01}c_{10}$, the matrix $C = \left(\begin{smallmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{smallmatrix}\right)$ has $\det C = 0$, so it has rank 1 and $C = \mathbf{a}\mathbf{b}^T$, i.e. $c_{jk} = a_j b_k$. **I is false:** for Φ^+, $c_{00}c_{11} = \frac{1}{2} \neq 0 = c_{01}c_{10}$. Answer: (D).

Paragraph for Questions 39 and 40

Hadamard test. A control qubit starts in $|0\rangle$ and a target register starts in an eigenstate $|u\rangle$ of a unitary U with $U\,|u\rangle = e^{2\pi i\varphi}\,|u\rangle$, $0 \le \varphi < 1$. The circuit applies, in order: (i) H on the control; (ii) controlled-U (control qubit $\rightarrow$ target register); (iii) H on the control; (iv) a measurement of the control qubit in the computational basis. This is the one-counting-qubit version of quantum phase estimation.

Q.39 Paragraph based · Hadamard test

The probability that the control qubit is found in state $|1\rangle$ is

(A) $\cos^2(\pi\varphi)$ **(B)** $\sin^2(\pi\varphi/2)$ **(C)** $\sin^2(2\pi\varphi)$ **(D)** ✓ $\sin^2(\pi\varphi)$

Answer: (D) Solution

After (i): $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\,|u\rangle$. After (ii), by phase kickback (Q.28 with $j = 0$): $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\varphi}\,|1\rangle\,)\,|u\rangle$. After (iii) (Q.23 with $\phi = 2\pi\varphi$):

$$\Big(\frac{1 + e^{2\pi i\varphi}}{2}\,|0\rangle + \frac{1 - e^{2\pi i\varphi}}{2}\,|1\rangle\Big)\,|u\rangle\,.$$

$|1 - e^{2\pi i\varphi}|^2 = 2 - 2\cos(2\pi\varphi) = 4\sin^2(\pi\varphi)$, so $P(1) = \sin^2(\pi\varphi)$ and $P(0) = \cos^2(\pi\varphi)$. $\varphi = 0 \Rightarrow$ always 0; $\varphi = \frac{1}{2} \Rightarrow$ always 1: one qubit resolves exactly one binary digit $\varphi = 0.\varphi_1$. More digits need more counting qubits and an inverse QFT.

Q.40 PARAGRAPH BASED · Repetition and confidence

Take $\varphi = \frac{1}{3}$. The circuit is run repeatedly (fresh $|0\rangle$ control each time). The minimum number of runs needed so that the probability of observing outcome 1 *at least once* exceeds 0.99 is

(A) 2 **(B)** 3 **(C)** ✓ 4 **(D)** 7

Answer: (C) Solution

From Q.39, $P(1) = \sin^2(\pi/3) = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}$, so a single run gives 0 with probability $\frac{1}{4}$. Runs are independent: $P(\text{no 1 in } k \text{ runs}) = \left(\frac{1}{4}\right)^k$. We need $1 - 4^{-k} > 0.99$, i.e. $4^k > 100$. $4^3 = 64 < 100$ (failure probability 0.0156), $4^4 = 256 > 100$ (failure probability 0.0039). Hence $k = 4$. (Repetition amplifies confidence; it does not make the single-shot estimate of $\varphi = \frac{1}{3} = 0.0101\ldots_2$ exact.)

SECTION C · QFT Basics Q.41–60

Q.41 Single-correct MCQ · QFT matrix element ($n = 2$)

For $n = 2$, the amplitude of $|10\rangle$ in QFT $|11\rangle$ is

(A) ✓ $-\frac{1}{2}$ **(B)** $\frac{1}{2}$ **(C)** $-\frac{i}{2}$ **(D)** $\frac{i}{2}$

Answer: (A) Solution

$N = 4$, $\omega = e^{2\pi i/4} = i$, $x = |11\rangle = 3$, $y = |10\rangle = 2$. From the definition

$$\langle y|\mathrm{QFT}|x\rangle = \frac{1}{\sqrt{N}}\,\omega^{xy} = \frac{1}{2}\,i^6 = \frac{1}{2}\,(i^4)(i^2) = -\frac{1}{2}.$$

Full output: QFT $|3\rangle = \frac{1}{2}(\,|0\rangle - i\,|1\rangle - |2\rangle + i\,|3\rangle\,)$, i.e. $\frac{1}{2}(|00\rangle - i\,|01\rangle - |10\rangle + i\,|11\rangle)$.

Q.42 Single-correct MCQ · Output distribution of QFT $|x\rangle$

A 5-qubit QFT is applied to the basis state $|x\rangle$ with $x = 19$ and the output is measured. The probability of obtaining $y = 7$ is

(A) $\frac{7}{32}$ **(B)** $\frac{1}{\sqrt{32}}$ **(C)** 0 **(D)** ✓ $\frac{1}{32}$

Answer: (D) Solution

QFT $|x\rangle = \frac{1}{\sqrt{32}}\sum_y \omega^{xy}\,|y\rangle$ and $|\omega^{xy}| = 1$, so every outcome has probability $\left|\frac{1}{\sqrt{32}}\right|^2 = \frac{1}{32}$, whatever x and y are. ($\frac{1}{\sqrt{32}}$ is the modulus of the amplitude, not a probability.) All the information about x sits in the phases ω^{19y}.

Q.43 Single-correct MCQ · Product formula ($n = 3$)

For $n = 3$, QFT $|110\rangle$ equals (qubit 1 written first)

(A) $\frac{1}{\sqrt{8}}(\,|0\rangle - |1\rangle\,)(\,|0\rangle - |1\rangle\,)(\,|0\rangle - i\,|1\rangle\,)$ **(B)** $\frac{1}{\sqrt{8}}(\,|0\rangle + |1\rangle\,)(\,|0\rangle - |1\rangle\,)(\,|0\rangle + i\,|1\rangle\,)$

(C) ✓ $\frac{1}{\sqrt{8}}(\,|0\rangle + |1\rangle\,)(\,|0\rangle - |1\rangle\,)(\,|0\rangle - i\,|1\rangle\,)$ **(D)** $\frac{1}{\sqrt{8}}(\,|0\rangle - i\,|1\rangle\,)(\,|0\rangle - |1\rangle\,)(\,|0\rangle + |1\rangle\,)$

Answer: (C) Solution

Product formula. Write $y = \sum_{l=1}^{n} y_l 2^{n-l}$; then $\omega^{xy} = \prod_l e^{2\pi i\, x y_l/2^l}$ and $e^{2\pi i\, x/2^l} = e^{2\pi i\, 0.x_{n-l+1}\ldots x_n}$ (only the last l bits of x survive), so the sum over y factorises into one factor per output qubit. For $n = 3$: QFT $|x_1x_2x_3\rangle = \frac{1}{\sqrt{8}}(\,|0\rangle + e^{2\pi i\, 0.x_3}\,|1\rangle\,)(\,|0\rangle + e^{2\pi i\, 0.x_2x_3}\,|1\rangle\,)(\,|0\rangle + e^{2\pi i\, 0.x_1x_2x_3}\,|1\rangle\,)$. With $x_1x_2x_3 = 110$: $0.x_3 = 0.0 = 0 \Rightarrow e^0 = 1$; $0.x_2x_3 = 0.10_2 = \frac{1}{2} \Rightarrow e^{i\pi} = -1$; $0.x_1x_2x_3 = 0.110_2 = \frac{3}{4} \Rightarrow e^{3\pi i/2} = -i$. Hence $\frac{1}{\sqrt{8}}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)(|0\rangle - i\,|1\rangle)$. *Check* with the definition: amplitude of y is $\omega^{6y}/\sqrt{8} = (-i)^y/\sqrt{8}$; for $y = 4y_1 + 2y_2 + y_3$, $(-i)^{4y_1}(-i)^{2y_2}(-i)^{y_3} = 1\cdot(-1)^{y_2}(-i)^{y_3}$ — the same three factors. The option $\frac{1}{\sqrt{8}}(|0\rangle - i\,|1\rangle)(|0\rangle - |1\rangle)(|0\rangle + |1\rangle)$ is the bit-reversed order (what you get without the final SWAPs).

Q.44 SINGLE-CORRECT MCQ · Controlled-rotation count

The textbook QFT circuit on $n = 12$ qubits contains how many controlled-R_k gates?

(A) 144 **(B)** 72 **(C)** ✓ 66 **(D)** 78

Answer: (C) **Solution**

Qubit j receives one controlled rotation from each qubit below it, i.e. $n - j$ of them. Total

$$\sum_{j=1}^{n}(n-j) = (n-1) + (n-2) + \cdots + 0 = \frac{n(n-1)}{2} = \frac{12 \cdot 11}{2} = 66.$$

($78 = \frac{n(n+1)}{2}$ also counts the 12 Hadamards; $144 = n^2$.)

Q.45 SINGLE-CORRECT MCQ · Two-qubit gates incl. SWAPs

On hardware a SWAP is implemented with 3 CNOTs. Counting every controlled-R_k as one two-qubit gate, the total number of two-qubit gates in the 12-qubit textbook QFT (including the final SWAPs) is

(A) 72 **(B)** ✓ 84 **(C)** 102 **(D)** 66

Answer: (B) **Solution**

Controlled rotations: $\frac{12\cdot 11}{2} = 66$ (Q.44). SWAPs: $\lfloor 12/2 \rfloor = 6$, each costing 3 CNOTs = 18. Total $66 + 18 = 84$. ($72 = 66 + 6$ counts each SWAP as one gate; $102 = 66 + 36$ uses 6 CNOTs per SWAP.) In practice the SWAPs are often removed by simply relabelling the output qubits, leaving 66.

Q.46 SINGLE-CORRECT MCQ · Smallest rotation angle

The smallest rotation angle appearing in the textbook QFT circuit on 6 qubits is

(A) 11.25° **(B)** 6° **(C)** 2.8125° **(D)** ✓ 5.625°

Answer: (D) **Solution**

The rotations used are $R_2, \ldots, R_n$ with angle $\frac{2\pi}{2^k}$. The smallest is $R_n = R_6$: $\frac{360^\circ}{2^6} = \frac{360^\circ}{64} = 5.625^\circ$. (11.25° is R_5; 2.8125° would be R_7, which needs 7 qubits.) These tiny angles are why the approximate QFT drops the smallest rotations (Section D).

Q.47 SINGLE-CORRECT MCQ · Phase on qubit 1; role of SWAPs

In the textbook QFT circuit acting on $|x_1 x_2 \ldots x_n\rangle$, *before* the final SWAPs, qubit 1 is in the state $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\,\phi}\,|1\rangle\,)$ with ϕ equal to

(A) ✓ $0.x_1x_2\ldots x_n$ **(B)** $0.x_nx_{n-1}\ldots x_1$ **(C)** $0.x_n$ **(D)** $0.x_1$

Answer: (A) **Solution**

H on qubit 1 gives $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.x_1}\,|1\rangle)$ because $(-1)^{x_1} = e^{2\pi i\, x_1/2}$. Each controlled-$R_k$ from qubit k (for $k = 2, \ldots, n$) multiplies the $|1\rangle$ part by $e^{2\pi i\, x_k/2^k}$, appending the binary digit x_k at position k: $0.x_1 \to 0.x_1x_2 \to \cdots \to 0.x_1x_2\ldots x_n$. This factor belongs to the *last* output qubit in the product formula, so the SWAPs move it from qubit 1 to qubit n.

Q.48 SINGLE-CORRECT MCQ · $\mathrm{QFT}^2 = \text{negation}$

For any n-qubit basis state, $\mathrm{QFT}^2 |x\rangle$ equals ($N = 2^n$)

(A) $|x\rangle$ **(B)** $|2x \bmod N\rangle$ **(C)** ✓ $|(-x) \bmod N\rangle$ **(D)** $|N-1-x\rangle$

Answer: (C) **Solution**

$$\mathrm{QFT}^2 |x\rangle = \frac{1}{N} \sum_z \Big(\sum_y \omega^{y(x+z)} \Big) |z\rangle .$$

The inner geometric sum equals N if $x + z \equiv 0 \pmod N$ and 0 otherwise. Hence $\mathrm{QFT}^2 |x\rangle = |-x \bmod N\rangle$. Examples ($n = 2$): $|0\rangle \to |0\rangle$, $|1\rangle \to |3\rangle$, $|2\rangle \to |2\rangle$. Consequently $\mathrm{QFT}^4 = I$. ($|N-1-x\rangle$ is the bitwise complement, a different map: it sends $|0\rangle \to |N-1\rangle$.)

Q.49 SINGLE-CORRECT MCQ · QFT of the uniform state

The QFT is applied to the uniform superposition $\frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$. The output is

(A) $|N-1\rangle$ **(B)** ✓ $|0\rangle$

(C) the uniform superposition itself **(D)** $|N/2\rangle$

Answer: (B) **Solution**

$$\frac{1}{\sqrt{N}} \sum_x \frac{1}{\sqrt{N}} \sum_y \omega^{xy} |y\rangle = \frac{1}{N} \sum_y \Big(\sum_x \omega^{xy} \Big) |y\rangle = |0\rangle ,$$

since $\sum_x \omega^{xy} = N \, \delta_{y,0}$. A constant ("DC") signal has only the zero-frequency component. Alternatively: the uniform superposition is $\mathrm{QFT}\,|0\rangle$ (product formula with $x = 0$), so applying the QFT once more gives $\mathrm{QFT}^2 |0\rangle = |-0 \bmod N\rangle = |0\rangle$ by Q.48.

Q.50 SINGLE-CORRECT MCQ · QFT of $|10\ldots0\rangle$

For n qubits, $\mathrm{QFT}\,|10\ldots0\rangle$ (i.e. $x = 2^{n-1}$) equals

(A) ✓ $|+\rangle^{\otimes(n-1)} \otimes |-\rangle$ **(B)** $|-\rangle^{\otimes n}$ **(C)** $|+\rangle^{\otimes n}$ **(D)** $|-\rangle \otimes |+\rangle^{\otimes(n-1)}$

Answer: (A) **Solution**

Amplitude of y: $\frac{1}{\sqrt{N}} \omega^{2^{n-1} y} = \frac{1}{\sqrt{N}} e^{2\pi i\, y/2} = \frac{1}{\sqrt{N}} (-1)^y$. The sign depends only on the parity of y, i.e. on its last bit y_n (qubit n). So the state factorises as $|+\rangle \otimes \cdots \otimes |+\rangle \otimes |-\rangle$. Product-formula check: $x_1 = 1$, others 0, so every factor has phase $e^{2\pi i\, 0.0\ldots} = 1$ except the last, $e^{2\pi i\, 0.x_1 \ldots x_n} = e^{2\pi i\, 0.1} = -1$. (This is the "highest frequency" alternating signal $+, -, +, -, \ldots$.)

Q.51 SINGLE-CORRECT MCQ · Constructing the inverse QFT

The inverse QFT on 3 qubits is built by reversing the textbook QFT circuit and conjugating every gate. Ignoring the SWAP, the *first* gate of the inverse circuit is

(A) ✓ H on qubit 3 **(B)** controlled-$R_3^\dagger$ between qubits 1 and 3

(C) controlled-$R_2^\dagger$ between qubits 1 and 2 **(D)** H on qubit 1

Answer: (A) **Solution**

$\mathrm{QFT} = \mathrm{SWAP}_{13} \cdot H_3 \cdot \mathrm{CR}_2^{(3\to2)} \cdot H_2 \cdot \mathrm{CR}_3^{(3\to1)} \cdot \mathrm{CR}_2^{(2\to1)} \cdot H_1$ (rightmost acts first). Since $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$:

$$\mathrm{QFT}^{-1} = H_1 \cdot \mathrm{CR}_2^{\dagger\,(2\to1)} \cdot \mathrm{CR}_3^{\dagger\,(3\to1)} \cdot H_2 \cdot \mathrm{CR}_2^{\dagger\,(3\to2)} \cdot H_3 \cdot \mathrm{SWAP}_{13},$$

so the order of application is SWAP, then H on qubit 3, then $CR_2^\dagger$ (3→2), H_2, …, finally H_1. $H^\dagger = H$ and $R_k^\dagger = \mathrm{diag}(1, e^{-2\pi i/2^k})$.

Q.52 SINGLE-CORRECT MCQ · Phase of a QFT amplitude

For $n = 4$, the phase (in degrees, in $[0°, 360°)$) of the amplitude of $|y\rangle$, $y = 3$, in QFT $|x\rangle$, $x = 5$, is

(A) 22.5° **(B)** ✓ 337.5° **(C)** 168.75° **(D)** 15°

Answer: (B) Solution

The amplitude is $\frac{1}{4}\omega^{xy}$ with $\omega = e^{2\pi i/16}$, so the phase is $\frac{2\pi\, xy}{16} = 2\pi \cdot \frac{15}{16}$, i.e. $\frac{15}{16} \times 360° = 337.5°$ (equivalently $-22.5°$; the option 22.5° has the sign convention reversed).

Q.53 SINGLE-CORRECT MCQ · QFT vs. NumPy FFT conventions

NumPy's two transforms of a length-N vector a are

$$\texttt{fft}(a)_y = \sum_x a_x\, e^{-2\pi i\, xy/N}, \qquad \texttt{ifft}(a)_y = \frac{1}{N}\sum_x a_x\, e^{+2\pi i\, xy/N}.$$

If QFT $\sum_x \alpha_x |x\rangle = \sum_y \beta_y |y\rangle$ (convention box), then the vector β equals

(A) $\texttt{fft}(\alpha)$ **(B)** ✓ $\sqrt{N} \cdot \texttt{ifft}(\alpha)$ **(C)** $\texttt{ifft}(\alpha)/\sqrt{N}$ **(D)** $\texttt{fft}(\alpha)/\sqrt{N}$

Answer: (B) Solution

$\beta_y = \frac{1}{\sqrt{N}} \sum_x \alpha_x e^{+2\pi i\, xy/N}$. The sign of the exponent is +, which matches `ifft`, and $\frac{1}{\sqrt{N}} = \sqrt{N} \cdot \frac{1}{N}$, so $\beta = \sqrt{N}\,\texttt{ifft}(\alpha)$ (`numpy.fft.ifft(alpha, norm="ortho")`). $\texttt{fft}(\alpha)/\sqrt{N}$ is the *inverse* QFT. This is how the course notebooks verify CUDA-Q's state vectors.

Q.54 SINGLE-CORRECT MCQ · QFT $|x\rangle$ is a product state

For an n-qubit computational basis state $|x\rangle$, the state QFT $|x\rangle$ is

(A) a product state only when x is even **(B)** always maximally entangled

(C) entangled unless $x = 0$ **(D)** ✓ always a product (unentangled) state

Answer: (D) Solution

The product formula writes QFT $|x\rangle$ as a tensor product of n single-qubit states $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.x_{n-l+1}\ldots x_n}|1\rangle)$, $l = 1, \ldots, n$. Hence it is never entangled, for *every* x. (The QFT *can* entangle superposition inputs, because the controlled rotations are entangling gates — compare Q.25.) This product structure is also why the textbook circuit needs only $O(n^2)$ gates.

Q.55 MATCHING LISTS · Gate count $n(n-1)/2$

Match the number of qubits n (List-I) with the number of controlled-R_k gates in the textbook QFT (List-II).

List-I		List-II	
(P)	$n = 4$	**(1)**	15
(Q)	$n = 6$	**(2)**	28
(R)	$n = 8$	**(3)**	45
(S)	$n = 10$	**(4)**	6

(A) P→1, Q→4, R→2, S→3
(B) P→4, Q→2, R→1, S→3
(C) ✓ P→4, Q→1, R→2, S→3
(D) P→4, Q→1, R→3, S→2

Answer: (C) **Solution**

Count $= \binom{n}{2} = \frac{n(n-1)}{2}$: $n = 4 \to 6$ (4), $n = 6 \to 15$ (1), $n = 8 \to 28$ (2), $n = 10 \to 45$ (3). Hence P→4, Q→1, R→2, S→3. Growth is quadratic, compared with the exponential 2^n amplitudes transformed.

Q.56 MATCHING LISTS · Two-qubit QFT outputs

For $n = 2$, match each input (List-I) with QFT of it (List-II); qubit 1 is written first.

List-I		List-II	
(P)	$\lvert 00\rangle$	**(1)**	$\frac{1}{2}(\lvert 0\rangle - \lvert 1\rangle)(\lvert 0\rangle + i\lvert 1\rangle)$
(Q)	$\lvert 01\rangle$	**(2)**	$\frac{1}{2}(\lvert 0\rangle - \lvert 1\rangle)(\lvert 0\rangle - i\lvert 1\rangle)$
(R)	$\lvert 10\rangle$	**(3)**	$\frac{1}{2}(\lvert 0\rangle + \lvert 1\rangle)(\lvert 0\rangle + \lvert 1\rangle)$
(S)	$\lvert 11\rangle$	**(4)**	$\frac{1}{2}(\lvert 0\rangle + \lvert 1\rangle)(\lvert 0\rangle - \lvert 1\rangle)$

(A) P→3, Q→4, R→1, S→2
(B) P→4, Q→1, R→3, S→2
(C) P→3, Q→2, R→4, S→1
(D) ✓ P→3, Q→1, R→4, S→2

Answer: (D) **Solution**

Product formula for $n = 2$: $\frac{1}{2}(\lvert 0\rangle + e^{2\pi i\, 0.x_2}\lvert 1\rangle)(\lvert 0\rangle + e^{2\pi i\, 0.x_1x_2}\lvert 1\rangle)$.

- 00: phases $e^0, e^0 \Rightarrow (|0\rangle + |1\rangle)(|0\rangle + |1\rangle)/2$ (3).
- 01: $0.x_2 = \frac{1}{2} \to -1$; $0.x_1x_2 = 0.01_2 = \frac{1}{4} \to i \Rightarrow$ (1).
- 10: $0.x_2 = 0 \to 1$; $0.10_2 = \frac{1}{2} \to -1 \Rightarrow$ (4).
- 11: $0.x_2 = \frac{1}{2} \to -1$; $0.11_2 = \frac{3}{4} \to -i \Rightarrow$ (2).

Hence P→3, Q→1, R→4, S→2. Check $|11\rangle$ against Q.41: expanding (2) gives $\frac{1}{2}(|00\rangle - i\,|01\rangle - |10\rangle + i\,|11\rangle)$, whose $|10\rangle$ amplitude is $-\frac{1}{2}$.

Q.57 Assertion–Reason · Why measuring QFT $|x\rangle$ reveals nothing

Assertion (A): Measuring QFT $|x\rangle$ in the computational basis reveals no information about the basis state x.

Reason (R): All N amplitudes of QFT $|x\rangle$ have the same magnitude $1/\sqrt{N}$.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Solution

R is true: $|\omega^{xy}/\sqrt{N}| = 1/\sqrt{N}$ for every y. **A is true** *because* of R: the outcome distribution $P(y) = 1/N$ is the same uniform distribution for every x, so the result is statistically independent of x. The information is entirely in the relative phases, which a computational-basis measurement discards; one must first undo the QFT (or interfere the phases in some other way) to read it. R is the correct explanation. Answer: (A).

Q.58 Statement I / II · What the QFT does *not* speed up

Statement I: The QFT on n qubits can be implemented with $O(n^2)$ one- and two-qubit gates.

Statement II: Consequently, for an arbitrary classical input vector of length 2^n, a quantum computer can output all 2^n of its Fourier coefficients using only $O(n^2)$ operations in total.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Solution

I is true: n Hadamards, $\frac{n(n-1)}{2}$ controlled rotations and $\lfloor n/2 \rfloor$ SWAPs. **II is false**, for two reasons: (1) *input* — loading an arbitrary 2^n-entry vector into amplitudes generally needs $\Omega(2^n)$ gates; (2) *output* — the coefficients are amplitudes; a measurement returns just one index y with probability $|\beta_y|^2$, and estimating all 2^n values needs exponentially many repetitions. The QFT is useful as a *subroutine* (phase estimation, Shor) where only a few peaks need to be read. Answer: (C).

Paragraph for Questions 59 and 60

The textbook 3-qubit QFT circuit is applied to the input $|x_1x_2x_3\rangle = |011\rangle$ ($x = 3$). The gates, in order, are: H on qubit 1; controlled-R_2 (control qubit 2, target qubit 1); controlled-R_3 (control qubit 3, target qubit 1); H on qubit 2; controlled-R_2 (control qubit 3, target qubit 2); H on qubit 3; SWAP of qubits 1 and 3. Here $R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$, and "the phase of qubit j" means the angle θ in its state $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\theta}|1\rangle)$, taken in $[0°, 360°)$.

Q.59 PARAGRAPH BASED · Circuit trace: phase build-up

Immediately after the controlled-R_3 gate (the third gate), the phase of qubit 1 is

(A) 270° **(B)** ✓ 135° **(C)** 90° **(D)** 45°

Answer: (B) Solution

Input $x_1 = 0$, $x_2 = 1$, $x_3 = 1$. H on qubit 1: phase $2\pi\, 0.x_1 = 0$, state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. C-$R_2$ (control $x_2 = 1$): adds $\frac{2\pi}{4} = 90°$. C-R_3 (control $x_3 = 1$): adds $\frac{2\pi}{8} = 45°$. Total $0° + 90° + 45° = 135°$, i.e. $2\pi \cdot 0.011_2 = 2\pi \cdot \frac{3}{8}$. (Since qubits 2 and 3 are basis states, the controls are classical here and nothing becomes entangled.)

Q.60 PARAGRAPH BASED · Circuit trace: final amplitude, SWAP

After the complete circuit (including the SWAP), the amplitude of the basis state $|110\rangle$ is

(A) $\frac{i}{8}$ **(B)** $\frac{e^{i\pi/4}}{\sqrt{8}}$ **(C)** $\frac{-i}{\sqrt{8}}$ **(D)** ✓ $\frac{i}{\sqrt{8}}$

Answer: (D) Solution

With the SWAP, the circuit implements exactly $\mathrm{QFT}\,|3\rangle = \frac{1}{\sqrt{8}}\sum_y \omega^{3y}|y\rangle$, $\omega = e^{i\pi/4}$. For $|110\rangle$, $y = 6$: $\omega^{18} = \omega^{16}\omega^2 = e^{i\pi/2} = i$, so the amplitude is $\frac{i}{\sqrt{8}}$. Gate-by-gate check: before the SWAP, qubit 1 has phase $\frac{3}{8}$ turn (Q.59), qubit 2 has $2\pi\, 0.x_2x_3 = 2\pi\, 0.11_2 = 270°$, qubit 3 has $2\pi\, 0.x_3 = 180°$. After swapping qubits 1 and 3, $|110\rangle$ collects phases $180° + 270° = 450° \equiv 90° \Rightarrow i$. Forgetting the SWAP gives $\frac{3}{8} + \frac{3}{4} = \frac{9}{8} \equiv \frac{1}{8}$ turn, i.e. $e^{i\pi/4}/\sqrt{8}$ — a wrong option.

SECTION D · Advanced QFT Q.61–80

Q.61 Single-correct MCQ · QFT of a periodic state $(r \mid N)$

On $n = 6$ qubits ($N = 64$) the state $\frac{1}{\sqrt{8}} \sum_{j=0}^{7} |3 + 8j\rangle$ is prepared, the QFT is applied and all qubits are measured. The possible outcomes y and their probabilities are

(A) only $y = 8$, with probability 1

(B) all 64 values of y, each with probability $\frac{1}{64}$

(C) $y \in \{3, 11, 19, \ldots, 59\}$, each with probability $\frac{1}{8}$

(D) ✓ $y \in \{0, 8, 16, \ldots, 56\}$, each with probability $\frac{1}{8}$

Answer: (D) Solution

With $\omega = e^{2\pi i/64}$,

$$\beta_y = \frac{1}{\sqrt{8}} \cdot \frac{1}{\sqrt{64}} \sum_{j=0}^{7} \omega^{(3+8j)y} = \frac{\omega^{3y}}{\sqrt{512}} \sum_{j=0}^{7} e^{2\pi i\, jy/8}.$$

The inner sum is 8 if $8 \mid y$ and 0 otherwise. Hence $\beta_y = \frac{8\,\omega^{3y}}{\sqrt{512}} = \frac{\omega^{3y}}{\sqrt{8}}$ for $y = 0, 8, \ldots, 56$ (8 values, probability $\frac{1}{8}$ each) and $\beta_y = 0$ elsewhere. A state with period r (here 8) dividing N produces r peaks at the multiples of N/r; the offset 3 only changes phases (see Q.63).

Q.62 Single-correct MCQ · Phases of QFT peaks

For the state $\frac{1}{\sqrt{8}} \sum_{j=0}^{7} |3 + 8j\rangle$ on 6 qubits, the phase (in $[0°, 360°)$) of the amplitude of $|16\rangle$ after the QFT is

(A) 90° **(B)** ✓ 270° **(C)** 135° **(D)** 0°

Answer: (B) Solution

From the Q.61 calculation, $\beta_{16} = \frac{1}{\sqrt{8}}\,\omega^{3\cdot 16} = \frac{1}{\sqrt{8}}\,e^{2\pi i\, 48/64} = \frac{1}{\sqrt{8}}\,e^{2\pi i \cdot \frac{3}{4}}$. Phase $= \frac{3}{4} \times 360° = 270°$ (the amplitude is $-\frac{i}{\sqrt{8}}$). The offset 3 is encoded in these phases: $\beta_{8k} \propto e^{2\pi i\, 3k/8}$, which is invisible to a single measurement but is what the shift theorem predicts.

Q.63 Single-correct MCQ · Shift theorem

Let $\mathrm{QFT} \sum_x \alpha_x |x\rangle = \sum_y \beta_y |y\rangle$ on n qubits ($N = 2^n$, $\omega = e^{2\pi i/N}$). For a fixed integer s, $\mathrm{QFT} \sum_x \alpha_x |(x + s) \bmod N\rangle$ equals

(A) $\sum_y \omega^{-sy} \beta_y |y\rangle$ **(B)** $\sum_y \beta_{(y+s) \bmod N} |y\rangle$

(C) $\sum_y \beta_y |(y + s) \bmod N\rangle$ **(D)** ✓ $\sum_y \omega^{sy} \beta_y |y\rangle$

Answer: (D) Solution

By linearity it is enough to look at one basis state: $\mathrm{QFT}\,|x + s\rangle = \frac{1}{\sqrt{N}} \sum_y \omega^{(x+s)y} |y\rangle = \sum_y \omega^{sy} \left(\frac{\omega^{xy}}{\sqrt{N}}\right) |y\rangle$ ($\omega^N = 1$ makes "mod N" harmless). Summing with weights α_x gives $\sum_y \omega^{sy} \beta_y |y\rangle$. **Shift theorem:** a shift in x becomes a phase ramp in y, and $|\beta_y|^2$ is unchanged. This is why the random, unknown offset x_0 left after measuring the work register in Shor's algorithm does not matter.

Q.64 Single-correct MCQ · QPE precision vs. counting qubits

To obtain φ accurate to n bits with probability at least $1-\varepsilon$, phase estimation uses $t = n + \lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$ counting qubits. For $n = 5$ and $\varepsilon = 0.1$, t equals

(A) 7 **(B)** ✓ 8 **(C)** 10 **(D)** 9

Answer: (B) Solution

$2 + \frac{1}{2\times 0.1} = 2 + 5 = 7$ and $\log_2 7 \approx 2.807$, so $\lceil \log_2 7 \rceil = 3$ and $t = 5 + 3 = 8$. The extra 3 qubits make the outcome distribution (whose peak has width about one grid point 2^{-t}, with tails $\propto 1/\delta^2$) concentrated enough that the first 5 bits are right with probability ≥ 0.9.

Q.65 Single-correct MCQ · QPE with exact t-bit phase

Phase estimation with $t = 3$ counting qubits (inverse QFT readout) is run on an eigenstate with $\varphi = \frac{5}{8}$. The probability that the counting register reads 101 is

(A) $\frac{5}{8}$ **(B)** ✓ 1 **(C)** $\frac{1}{8}$ **(D)** $\frac{4}{\pi^2}$

Answer: (B) Solution

After the controlled-U^{2^j} gates the counting register holds $\frac{1}{\sqrt{8}} \sum_{k=0}^{7} e^{2\pi i \varphi k} |k\rangle = \frac{1}{\sqrt{8}} \sum_k \omega^{5k} |k\rangle = \mathrm{QFT}\,|5\rangle$ ($\omega = e^{2\pi i/8}$), because $\varphi = \frac{5}{8} = 0.101_2$ is exactly representable with 3 bits. The inverse QFT returns $|5\rangle = |101\rangle$ with certainty. The bound $\frac{4}{\pi^2}$ applies only when φ is *not* a multiple of 2^{-t}.

Q.66 Single-correct MCQ · The $4/\pi^2$ success bound

If φ is not exactly a t-bit binary fraction, phase estimation returns the *single* closest t-bit approximation with probability at least

(A) $\frac{1}{2}$ **(B)** $\frac{1}{\pi} \approx 0.318$ **(C)** ✓ $\frac{4}{\pi^2} \approx 0.405$ **(D)** $\frac{8}{\pi^2} \approx 0.81$

Answer: (C) Solution

Let b be the closest grid point and $\delta = \varphi - b/2^t$, with $0 < |\delta| \leq 2^{-(t+1)}$. The amplitude of b is

$$\alpha_b = \frac{1}{2^t} \sum_{k=0}^{2^t-1} e^{2\pi i k \delta}, \qquad |\alpha_b| = \frac{|\sin(\pi 2^t \delta)|}{2^t |\sin(\pi \delta)|}.$$

Since $\pi 2^t |\delta| \leq \frac{\pi}{2}$, use $\sin x \geq \frac{2x}{\pi}$ there (concavity) in the numerator and $\sin x \leq x$ in the denominator: $|\alpha_b| \geq \frac{2 \cdot 2^t |\delta|}{2^t \pi |\delta|} = \frac{2}{\pi}$, so $P(b) \geq \frac{4}{\pi^2} \approx 0.405$. ($\frac{8}{\pi^2}$ is the bound for getting *one of the two* nearest grid points.)

Q.67 Single-correct MCQ · QPE distribution for $\varphi = 1/3$

Phase estimation with $t = 2$ counting qubits is run with $\varphi = \frac{1}{3}$. The probability of the outcome $y = 1$ (i.e. the estimate $\varphi \approx \frac{1}{4}$) is

(A) $\dfrac{6 - 3\sqrt{3}}{16}$ **(B)** $\dfrac{1}{3}$ **(C)** $\dfrac{4}{\pi^2}$ **(D)** ✓ $\dfrac{6 + 3\sqrt{3}}{16}$

Answer: (D) Solution

$P(y) = \frac{1}{16}\left|\sum_{k=0}^{3} e^{2\pi i k\delta}\right|^2 = \frac{1}{16}\frac{\sin^2(4\pi\delta)}{\sin^2(\pi\delta)}$ with $\delta = \varphi - \frac{y}{4}$. For $y = 1$: $\delta = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$, so

$$P(1) = \frac{1}{16}\frac{\sin^2(\pi/3)}{\sin^2(\pi/12)} = \frac{1}{16}\cdot\frac{3/4}{(1-\cos\frac{\pi}{6})/2} = \frac{1}{16}\cdot\frac{3}{2-\sqrt{3}} = \frac{3(2+\sqrt{3})}{16} = \frac{6+3\sqrt{3}}{16} \approx 0.700.$$

For completeness $P(0) = \frac{1}{16}$, $P(2) = \frac{3}{16}$, $P(3) \approx 0.050$; the four add to 1. It exceeds the guaranteed $\frac{4}{\pi^2} \approx 0.405$ because $\frac{1}{3}$ is not at the worst position (midway between grid points).

Q.68 Single-correct MCQ · Qubit budget of Shor's algorithm

For Shor's algorithm with a work register of $n = \lceil \log_2 N \rceil$ qubits and a counting register of $2n$ qubits, the total number of qubits needed to factor $N = 21$ is

(A) ✓ 15 **(B)** 12 **(C)** 21 **(D)** 10

Answer: (A) Solution

$\log_2 21 \approx 4.39$, so $n = 5$ (5 bits hold the residues $0, \ldots, 20$). Counting register $2n = 10$ qubits, work register 5 qubits: total 15. The $2n$ counting qubits make $M = 2^{2n} \geq N^2$, which guarantees that each peak y/M is within $\frac{1}{2r^2}$ of some k/r, so continued fractions can recover r. (The passage uses exactly these $10 + 5$ qubits.)

Q.69 Single-correct MCQ · Order finding, $a = 7$, $N = 15$

Order finding for $a = 7$, $N = 15$ uses a counting register of $t = 8$ qubits ($M = 256$). The set of possible measurement outcomes y is

(A) $\{0, 128\}$ **(B)** ✓ $\{0, 64, 128, 192\}$ **(C)** $\{0, 4, 8, 12\}$ **(D)** $\{0, 85, 171\}$

Answer: (B) Solution

Powers of 7 mod 15: $7, 49 \equiv 4, 28 \equiv 13, 91 \equiv 1$, so $r = 4$. After measuring (or ignoring) the work register, the counting register is periodic with period 4: $\frac{1}{8}\sum_{j=0}^{63}|x_0 + 4j\rangle$. Since $4 \mid 256$, by Q.61 the outcomes are exactly the multiples of $M/r = 64$: $y \in \{0, 64, 128, 192\}$, each with probability $\frac{1}{4}$, with no leakage. ($\{0, 85, 171\}$ would correspond to $r = 3$.)

Q.70 Single-correct MCQ · Failure mode $\gcd(k, r) > 1$

In the setting of the previous question (order finding for $a = 7$, $N = 15$, $M = 256$), the continued-fraction step turns $y/256$ into k/r'. Which outcome leads to the *wrong* candidate $r' = 2$?

(A) $y = 64$ **(B)** ✓ $y = 128$ **(C)** $y = 0$ **(D)** $y = 192$

Answer: (B) **Solution**

$\frac{64}{256} = \frac{1}{4} \Rightarrow r' = 4$ ✓; $\frac{192}{256} = \frac{3}{4} \Rightarrow r' = 4$ ✓; $\frac{128}{256} = \frac{1}{2} \Rightarrow r' = 2$; $\frac{0}{256} = \frac{0}{1} \Rightarrow r' = 1$ (no information). For $y = 128$, $k = 2$ shares the factor 2 with $r = 4$, so the fraction reduces and only a *divisor* of r appears. The check $7^2 = 49 \equiv 4 \not\equiv 1 \pmod{15}$ rejects it; one repeats the run or takes the lcm of several candidates. With the correct $r = 4$: $\gcd(7^2 - 1, 15) = \gcd(48, 15) = 3$, $\gcd(7^2 + 1, 15) = \gcd(50, 15) = 5$.

Q.71 Single-correct MCQ · Approximate QFT gate savings

An approximate QFT on $n = 20$ qubits keeps only the controlled-R_k gates with $k \le m = 6$ and drops the rest. The number of controlled rotations dropped is

(A) 114 **(B)** 85 **(C)** 95 **(D)** ✓ 105

Answer: (D) **Solution**

In the textbook circuit the gate R_k (for $k = 2, \dots, n$) appears $n - k + 1$ times (once for each target qubit j with $j + k - 1 \le n$). Total $= \binom{20}{2} = 190$. Kept: $k = 2, \dots, 6$: $19 + 18 + 17 + 16 + 15 = 85$. Dropped: $190 - 85 = 105$. The approximate QFT therefore needs only $O(nm)$ gates instead of $O(n^2)$, at the price of a small error (Q.77).

Q.72 Single-correct MCQ · Periodic state with $r \nmid N$

On $n = 6$ qubits ($N = 64$) the uniform superposition of all $x \equiv 0 \pmod 6$, $0 \le x \le 63$, is prepared and the QFT applied. The probability of the outcome $y = 32$ is

(A) ✓ $\frac{11}{64}$ **(B)** $\frac{1}{64}$ **(C)** $\frac{1}{6}$ **(D)** $\frac{1}{11}$

Answer: (A) **Solution**

The allowed x are $0, 6, 12, \dots, 60$: $K = 11$ terms ($6 \times 10 = 60 \le 63 < 66$). Then

$$\beta_{32} = \frac{1}{\sqrt{11}} \cdot \frac{1}{\sqrt{64}} \sum_{j=0}^{10} e^{2\pi i\, 6j \cdot 32/64} = \frac{1}{\sqrt{704}} \sum_{j=0}^{10} e^{2\pi i \cdot 3j} = \frac{11}{\sqrt{704}},$$

so $P(32) = \frac{121}{704} = \frac{11}{64} \approx 0.172$. The period 6 does not divide 64, so the ideal answer $\frac{1}{6}$ is only approximate, and most peaks ($y \approx 10.67k$) are smeared over neighbouring integers; $y = 32 = 3 \cdot \frac{64}{6}$ happens to fall exactly on a grid point.

Q.73 Single-correct MCQ · Eigenvalues of the QFT

Every eigenvalue λ of the n-qubit QFT matrix ($n \geq 1$) satisfies

(A) $\lambda^N = 1$, and all N of the N-th roots of unity occur

(B) $|\lambda| < 1$ for at least one eigenvalue

(C) $\lambda \in \{1, -1\}$

(D) ✓ $\lambda \in \{1, -1, i, -i\}$

Answer: (D) Solution

From Q.48, $\mathrm{QFT}^2 |x\rangle = |-x\rangle$, hence $\mathrm{QFT}^4 = I$. If $F\mathbf{v} = \lambda\mathbf{v}$ then $\mathbf{v} = F^4\mathbf{v} = \lambda^4\mathbf{v}$, so $\lambda^4 = 1$: $\lambda \in \{1, -1, i, -i\}$. The QFT is unitary, so $|\lambda| = 1$ always. Imaginary eigenvalues do occur (e.g. i for $N = 4$, where $\operatorname{tr} F = 1 + i$), so $\{1, -1\}$ is too small; for $N \geq 8$ all four values occur, but never all N roots of unity once $N > 4$.

Q.74 Single-correct MCQ · Semiclassical QFT

In phase estimation the inverse QFT is immediately followed by measurement of every counting qubit. Using the semiclassical (Griffiths–Niu) QFT, the number of *two-qubit* gates needed for this inverse QFT on n qubits is

(A) $n - 1$ **(B)** $\dfrac{n(n-1)}{2}$ **(C)** ✓ 0 **(D)** n

Answer: (C) Solution

Controlled-phase gates are symmetric (Q.27), so in each $\mathrm{C}R_k^\dagger$ the qubit that is measured *first* can be taken as the control. Once a qubit has been measured, its value y_j is a classical bit, and "controlled on a measured qubit" becomes "apply $R_k^\dagger$ if $y_j = 1$": a classically controlled single-qubit gate. Measuring the qubits one after the other and feeding each result forward removes all $\frac{n(n-1)}{2}$ two-qubit gates. This needs only mid-circuit measurement plus feed-forward, and even allows one counting qubit to be recycled n times.

Q.75 Matching lists · Applications of the QFT

Match each algorithm in List-I with the role played by the QFT, or the related step, in List-II.

List-I		List-II	
(P)	Quantum phase estimation	**(1)**	adds a number by applying controlled phase rotations to a register held in the Fourier basis
(Q)	Shor's factoring algorithm	**(2)**	turns the phases $e^{2\pi i\, 2^j \varphi}$ in the counting register into the binary digits of φ
(R)	Draper's quantum adder	**(3)**	after a QPE step, applies an ancilla rotation proportional to $1/\lambda$ for each eigenvalue λ
(S)	HHL linear-system solver	**(4)**	extracts an order r from peaks near kM/r using continued fractions

(A) ✓ P→2, Q→4, R→1, S→3 **(B)** P→4, Q→2, R→1, S→3

(C) P→2, Q→1, R→4, S→3 **(D)** P→2, Q→4, R→3, S→1

Answer: (A) Solution

QPE: controlled-U^{2^j} writes $\frac{1}{\sqrt{2^t}}\sum_k e^{2\pi i\varphi k}|k\rangle \approx \mathrm{QFT}\,|2^t\varphi\rangle$; the inverse QFT reads the digits (2). Shor: QPE applied to modular multiplication, whose eigenphases are k/r; continued fractions give r (4). Draper: in the Fourier basis, adding a to $|x\rangle$ is just the phase ramp ω^{ay} (Q.63), done with single-qubit or controlled rotations, then an inverse QFT (1). HHL: QPE puts eigenvalues λ_j in a register, a controlled rotation of an ancilla by an angle with $\sin\theta \propto 1/\lambda_j$ performs A^{-1}, then QPE is uncomputed (3). Hence P→2, Q→4, R→1, S→3.

Q.76 Matching lists · Period ↔ peak spacing

On $n = 4$ qubits ($N = 16$) the QFT is applied to each input in List-I and the result measured. Match with List-II.

List-I		List-II	
(P)	$\|0\rangle$	**(1)**	only $y = 0$ and $y = 8$, each with probability $\frac{1}{2}$
(Q)	$\frac{1}{2}(\,\|0\rangle + \|4\rangle + \|8\rangle + \|12\rangle\,)$	**(2)**	all 16 outcomes, each with probability $\frac{1}{16}$
(R)	$\frac{1}{\sqrt{8}}\sum_{x\ \text{even}}\|x\rangle$	**(3)**	only $y = 0, 4, 8, 12$, each with probability $\frac{1}{4}$
(S)	$\frac{1}{4}\sum_{x=0}^{15}\|x\rangle$	**(4)**	$y = 0$ with certainty

(A) P→4, Q→3, R→1, S→2
(B) P→2, Q→3, R→4, S→1
(C) ✓ P→2, Q→3, R→1, S→4
(D) P→2, Q→1, R→3, S→4

Answer: (C) Solution

P: a basis state transforms to equal-magnitude amplitudes (Q.42): uniform over 16 (2). Q: period $r = 4$ dividing 16: peaks at multiples of $16/4 = 4$, i.e. $0, 4, 8, 12$, each $\frac{1}{4}$ (3). R: the even numbers have period $r = 2$: peaks at multiples of $16/2 = 8$, i.e. 0 and 8, each $\frac{1}{2}$ (1). S: the uniform superposition has period 1 and goes to $|0\rangle$ (Q.49) (4). Pattern: period r in $x \Leftrightarrow r$ peaks spaced N/r in y — the "uncertainty principle" of the Fourier transform. Hence P→2, Q→3, R→1, S→4.

Q.77 Assertion–Reason · Approximate QFT error bound

Assertion (A): For large n, dropping every controlled-R_k with $k > m$, where m is only slightly larger than $\log_2 n$, changes the output of the QFT circuit only slightly.

Reason (R): Each dropped gate differs from the identity only by a phase $2\pi/2^k$, so its error is at most about $2\pi/2^k$; at most n gates are dropped for each k, giving a total error of order $n \cdot 2\pi/2^m$, which is small once $2^m \gg n$.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Answer: (A) Solution

R is true: $\|CR_k - I\| = |e^{2\pi i/2^k} - 1| = 2\sin(\pi/2^k) \le 2\pi/2^k$. Errors of a product of unitaries add at most linearly (triangle inequality), and R_k appears $n - k + 1 \le n$ times, so the total error is at most $\sum_{k>m} n \cdot \frac{2\pi}{2^k} < \frac{2\pi n}{2^m}$. **A is true** *because* of R: choosing $m = \log_2 n + c$ makes the error at most $2\pi/2^c$, independent of n. (Coppersmith's approximate QFT; in practice gate noise is larger than the error from these tiny rotations, so dropping them even *improves* real results.) Answer: (A).

Q.78 Statement I / II · Non-terminating binary phases

Statement I: For $\varphi = \frac{1}{3}$, phase estimation with any finite number t of counting qubits never returns an outcome y with $y/2^t = \varphi$ exactly, with certainty or otherwise.

Statement II: The number $\frac{1}{3}$ has no terminating binary expansion ($\frac{1}{3} = 0.010101\ldots_2$).

(A) ✓ Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (A) Solution

II is true: $\frac{1}{3} = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots = 0.\overline{01}_2$; a terminating expansion would mean $\frac{1}{3} = \frac{y}{2^t}$, i.e. $2^t = 3y$, impossible since $3 \nmid 2^t$. **I is true:** the outcome is always an integer y, so the estimate is $y/2^t \neq \frac{1}{3}$ for every t by the same argument. The output distribution is peaked near $2^t/3$ (see Q.67: $t = 2$ gives $y = 1$ with probability ≈ 0.70). Both statements are true. Answer: (A).

Paragraph for Questions 79 and 80

Shor's algorithm for $N = 21$ with $a = 2$. A counting register of $t = 10$ qubits ($M = 2^{10} = 1024$) and a 5-qubit work register initialised to $|1\rangle$ are used. Steps: (i) $H^{\otimes 10}$ on the counting register; (ii) $|x\rangle\,|1\rangle \mapsto |x\rangle\,|2^x \bmod 21\rangle$; (iii) inverse QFT on the counting register; (iv) measure the counting register, obtaining y. Classical post-processing replaces $y/1024$ by the closest fraction k/r with $r < 21$ (continued fractions) and keeps r if $2^r \equiv 1 \pmod{21}$. The true order of 2 modulo 21 is 6, since $2^6 = 64 = 3 \cdot 21 + 1$.

Q.79 Paragraph based · Shor, $N = 21$: continued fractions

Which of the following measured values y leads the post-processing to the correct order $r = 6$?

(A) 683 **(B)** 512 **(C)** ✓ 171 **(D)** 341

Answer: (C) Solution

The peaks lie near $y \approx k \cdot \frac{1024}{6} = 170.67k$: $0, 170.67, 341.33, 512, 682.67, 853.33$.

- $\frac{171}{1024} = 0.16699$: continued fraction $[0; 5, 1, 84, 2]$, convergents $\frac{1}{5}, \frac{1}{6}, \frac{85}{509}, \frac{171}{1024}$; the last with denominator < 21 is $\frac{1}{6} \Rightarrow r = 6$, and $2^6 \equiv 1$ ✓.
- $\frac{341}{1024} \approx \frac{1}{3}$ ($k = 2$), $\frac{512}{1024} = \frac{1}{2}$ ($k = 3$), $\frac{683}{1024} \approx \frac{2}{3}$ ($k = 4$): $\gcd(k, 6) > 1$, so only the divisors $3, 2, 3$ appear and $2^3 = 8$, $2^2 = 4$ are $\not\equiv 1 \pmod{21}$.

Factoring: $2^{r/2} = 8$, $\gcd(8 - 1, 21) = 7$ and $\gcd(8 + 1, 21) = 3$; indeed $21 = 3 \times 7$. Only $k = 1, 5$ (coprime to 6) give r directly, i.e. about $\frac{\varphi(6)}{6} = \frac{1}{3}$ of the runs.

Q.80 PARAGRAPH BASED · Shor, $N = 21$: peak probability

The probability that the measured value is exactly $y = 512$ is closest to

(A) 0.333 **(B)** 0.500 **(C)** ✓ 0.167 **(D)** 0.001

Answer: (C) Solution

After step (ii) the state is $\frac{1}{\sqrt{1024}} \sum_{x=0}^{1023} |x\rangle\,|2^x \bmod 21\rangle$. Group $x = x_0 + 6j$ with $x_0 \in \{0, \ldots, 5\}$; because $1024 = 6 \cdot 170 + 4$, the offsets $x_0 = 0, 1, 2, 3$ have $K = 171$ terms and $x_0 = 4, 5$ have $K = 170$ terms (check: $4 \cdot 171 + 2 \cdot 170 = 1024$). The six work values $2^{x_0} \bmod 21$ are distinct, so the branches do not interfere. After the inverse QFT, the amplitude of $|y\rangle$ in branch x_0 is $\frac{1}{1024} \sum_j e^{-2\pi i (x_0 + 6j) y / 1024}$. For $y = 512$, $e^{-2\pi i \cdot 6j \cdot 512/1024} = e^{-6\pi i j} = 1$, so its modulus is $K_{x_0}/1024$ and

$$P(512) = \frac{4 \cdot 171^2 + 2 \cdot 170^2}{1024^2} = \frac{174\,764}{1\,048\,576} \approx 0.16667 \approx \frac{1}{6}.$$

$y = 512 = 3 \cdot \frac{1024}{6}$ falls exactly on a grid point, so this peak is "perfect"; the peaks at $k = 1, 2, 4, 5$ are smeared over neighbouring y (the nearest integers $171, 341, 683, 853$ get only about 0.11 each).